

Intel® SS72 NAND Flash Memory

JS29F02G08AANB3, JS29F04G08BANB3, JS29F08G08FANB3

Datasheet

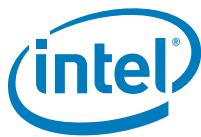
Product Features

- **Organization:**
 - Page size: x8: 2,112 bytes (2,048 + 64 bytes); x16: 1,056 words (1,024 + 32 words)
 - Block size: 64 pages (128K + 4K bytes)
 - Device size: 2Gb: 2,048 blocks; 4Gb: 4,096 blocks; 8Gb: 8,192 blocks
- **Read performance:**
 - Random read: 25µs
 - Sequential read: 30ns (3V x8 only)
- **Write performance:**
 - Page program: 300µs (TYP)
 - Block erase: 2ms (TYP)
- **Endurance:**
 - 100,000 PROGRAM/ERASE cycles
 - Data retention: 10 years
- **First block:**
 - Block address 00h: guaranteed valid without ECC (up to 1,000 PROGRAM/ERASE cycles)
- **VCC:**
 - 1.70V–1.95V
 - 2.7V–3.6V
- **Automated PROGRAM and ERASE**
- **Basic NAND flash command set:**
 - PAGE READ, READ for INTERNAL DATA MOVE, RANDOM DATA READ, READ ID, READ STATUS, PROGRAM PAGE, RANDOM DATA INPUT, PROGRAM PAGE CACHE MODE, PROGRAM for INTERNAL DATA MOVE, BLOCK ERASE, RESET
- **New commands:**
 - PAGE READ CACHE MODE
 - One-time programmable (OTP), including: OTP DATA PROGRAM, OTP DATA PROTECT, OTP DATA READ
 - READ UNIQUE ID (contact factory)
 - READ ID2 (contact factory)
- **Ready/busy# (R/B#) Pin**
 - Provides a hardware method for detecting PROGRAM or ERASE cycle completion
- **WP# Signal:**
 - Hardware write protect
- **Operating Temperature:**
 - Commercial (0 °C to 70 °C)
 - Extended (–40 °C to +85 °C)

Intel® NAND Flash technology provides a cost-effective solution for applications requiring high-density solid-state storage. Intel® JS29F02G08AANB3 devices are 2Gb NAND Flash memory devices. The JS29F04G08BANB3 devices are 4Gb devices. The JS29F08G08FANB3 devices are four-die stacks that operate as two independent 4Gb devices, providing a total storage capacity of 8Gb in a single, space-saving package. These devices include standard NAND features as well as new features designed to enhance system-level performance.

Intel® NAND Flash devices use a highly multiplexed 8- or 16-bit bus (I/O[7:0] or I/O[15:0]) to transfer data, addresses, and instructions. The five command signals (CLE, ALE, CE#, RE#, WE#) implement the NAND command bus interface protocol. Two additional signals control hardware write protection (WP#), and monitor device status (R/B#).

This hardware interface creates a low-signal-count device with a standard interface that is the same from one density to another, supporting future upgrades to higher densities without board redesign. Intel® JS29F02G08AANB3 and JS29F04G08BANB3 devices contain 2,048 and 4,096 erasable blocks respectively. Each block is subdivided into 64 programmable pages. Each page consists of 2,112 bytes



(x8) or 1,056 words (x16). The pages are further divided into a 2,048-byte data storage region with a separate 64-byte area on the x8 device; and on the x16 device, separate 1,024-word and 32-word areas. The 64-byte and 32-word areas are typically used for error management functions.

The contents of each 2,112-byte page can be programmed in 300 μ s, and an entire 132K-byte/66K-word block can be erased in 2ms. On-chip control logic automates PROGRAM and ERASE operations to maximize cycle endurance. ERASE/PROGRAM endurance is specified at 100K cycles when using appropriate error correcting code (ECC) and error management.

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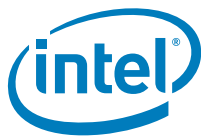
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Revision History

Revision Date	Revision	Description
29-Aug-06	006	- First production datasheet revision - Removed the PRE signal due to low customer demand. - Clarified in Figure 6 "48-Pin TSOP Type 1 Pin Assignment Top View" on page 13 that pin 38 can be either DNU or Vss.
18-Aug-06	005	Updated with new product naming convention, and document title change.
29-Jun-06	004	- Added x16 page size content throughout document. - Added Operating Temperature to the Product Features section, on page 1.
3-May-06	003	Updated graphics and ordering information.
31-Mar-06	002	revised for 4 and 8 Gbit parts.
10-Mar-06	001	Initial Release.



1.0 Functional Overview

This section provides an overview of the device in the following sections:

- [Section 1.1, "Architecture" on page 7](#)
- [Section 1.2, "Memory Map and Addressing" on page 8](#)

This section provides an overview of the device architecture, addressing, and memory map.

1.1 Architecture

These devices use NAND Flash electrical and command interfaces. Data, commands, and addresses are multiplexed onto the same pins. This provides a memory device with a low signal count.

The internal memory array is accessed on a page basis. When doing reads, a page of data is copied from the memory array into the data register. Once copied to the data register, data is output sequentially, byte by byte on x8 devices, or word by word on x16 devices.

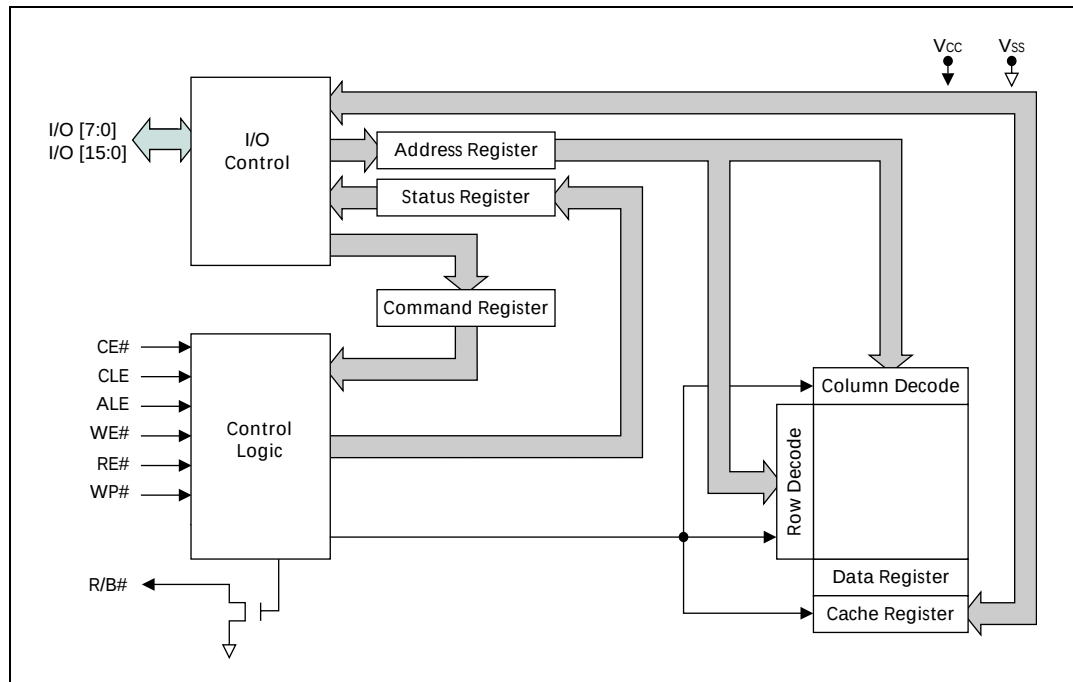
The memory array is programmed on a page basis. After the starting address is loaded into the internal address register, data is sequentially written to the internal data register up to the end of a page. After all of the page data has been loaded into the data register, array programming is started.

In order to increase programming bandwidth, this device incorporates a cache register. In the cache programming mode, data is first copied into the cache register and then into the data register. Once the data is copied into the data register, programming begins.

After the data register has been loaded and programming started, the cache register becomes available for loading additional data. Loading the next page of data into the cache register takes place while page programming is in process.

The INTERNAL DATA MOVE command also uses the internal cache register. Normally, moving data from one area of external memory to another uses a large number of external memory cycles. By using the internal cache register and data register, array data can be copied from one page and then programmed into another without using external memory cycles.

Figure 1. Intel® SS72 NAND Flash Memory Functional Block Diagram



1.2 Memory Map and Addressing

NAND Flash devices do not contain dedicated address signals. Addresses are loaded using a five-cycle sequence as shown in [Table 2, "Intel® SS72 NAND Flash Memory Array Addressing \(x8\)"](#) on page 10 for the x8 device and in [Table 4, "Intel® SS72 NAND Flash Memory Array Addressing \(x16\)"](#) on page 12 for the x16 device.



1.2.1 Memory Map and Array Addressing (x8 Device)

The 12-bit column address is capable of addressing from 0 to 4,095 words on x8 devices; however, only words 0 through 2,111 are valid. Words 2,112 through 4,095 of each page are “out of bounds,” do not exist in the device, and cannot be addressed.

Figure 2. Intel® SS72 NAND Flash Memory Memory Map (x8)

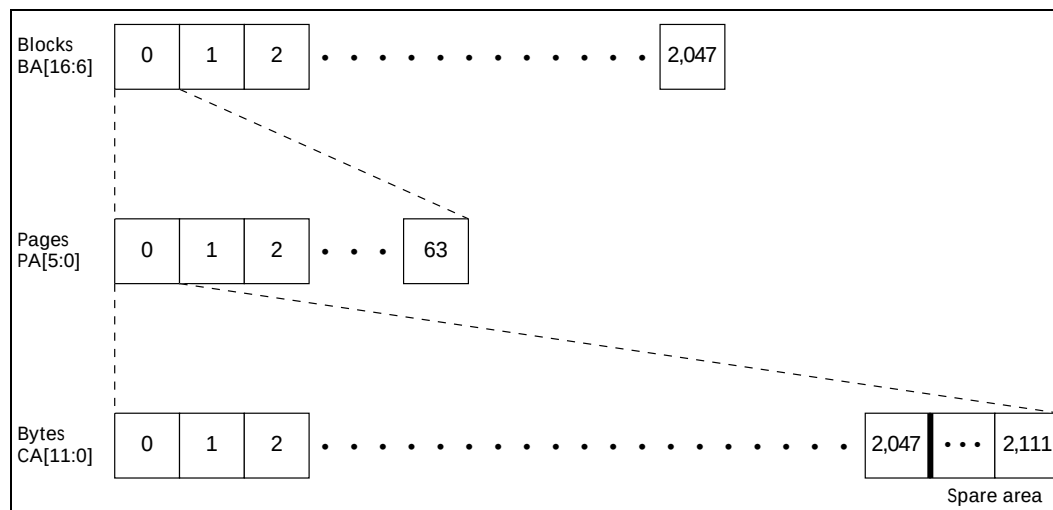


Table 1. Intel® SS72 NAND Flash Memory Operational Example (x8)

Block	Page	Min Address in Page	Max Address in Page	Out of Bounds Addresses in Page
0	0	0x000000000	0x000000083F	0x0000000840-0x0000000FFF
0	1	0x0000010000	0x000001083F	0x0000010840-0x0000010FFF
0	2	0x0000020000	0x000002083F	0x0000020840-0x0000020FFF
...	...	...	...	...
2,046	62	0x01FFFE0000	0x01FFFE083F	0x01FFFE0840-0x01FFFE0FFF
2,047	63	0x01FFFF0000	0x01FFFF083F	0x01FFFF0840-0x01FFFF0FFF

Note: As shown in Table 2, “Intel® SS72 NAND Flash Memory Array Addressing (x8)” on page 10, the high nibble of ADDRESS cycle 2 has no assigned address bits; however, these 4 bits must be held LOW during the ADDRESS cycle to ensure that the address is interpreted correctly by the NAND Flash device. These extra bits are accounted for in ADDRESS cycle 2 even though they do not have address bits assigned to them.

Figure 3. Intel® SS72 NAND Flash Memory Array Organization (x8)

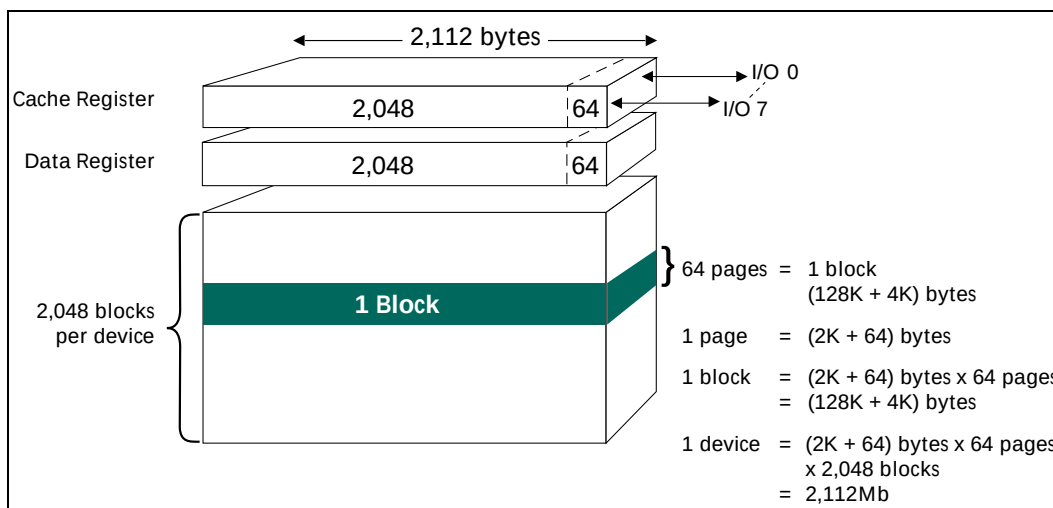


Table 2. Intel® SS72 NAND Flash Memory Array Addressing (x8)

Cycle	I/O7	I/O6	I/O5	I/O4	I/O3	I/O2	I/O1	I/O0
First	CA7	CA6	CA5	CA4	CA3	CA2	CA1	CA0
Second	LOW	LOW	LOW	LOW	CA11 ²	CA10	CA9	CA8
Third	BA7	BA6	PA5	PA4	PA3	PA2	PA1	PA0
Fourth	BA15	BA14	BA13	BA12	BA11	BA10	BA9	BA8
Fifth	LOW	LOW	LOW	LOW	LOW	LOW	LOW	BA16

Notes:

- Block address concatenated with page address = actual page address. CAx = column address; PAx = page address, BAx = block address.
- If CA11 = "1" then CA[10:6] must be "0."



1.2.2 Memory Map and Array Addressing (x16 Device)

The 11-bit column address is capable of addressing from 0 to 2,047 words on x16 devices; however, only words 0 through 1,055 are valid. Words 1,056 through 2,048 of each page are "out of bounds," do not exist in the device, and cannot be addressed.

Figure 4. Intel® SS72 NAND Flash Memory Memory Map (x16)

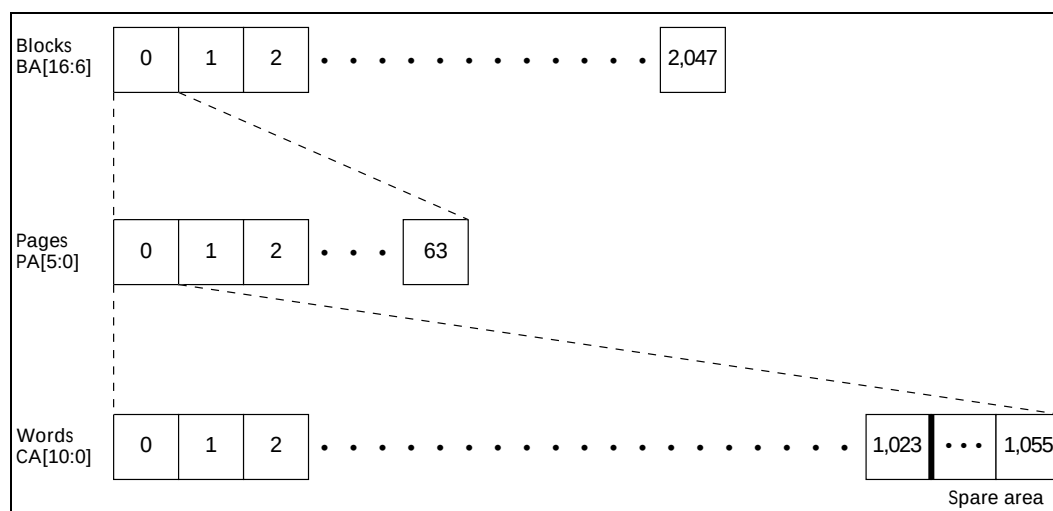


Table 3. Intel® SS72 NAND Flash Memory Operational Example (x16)

Block	Page	Min Address in Page	Max Address in Page	Out of Bounds Addresses in Page
0	0	0x000000000	0x000000041F	0x0000000420-0x0000000FFF
0	1	0x0000010000	0x000001041F	0x0000010420-0x0000010FFF
0	2	0x0000020000	0x000002041F	0x0000020420-0x0000020FFF
...	...	...	...	...
2,046	62	0x01FFFE0000	0x01FFFE041F	0x01FFFE0420-0x01FFFE0FFF
2,047	63	0x01FFFF0000	0x01FFFF041F	0x01FFFF0420-0x01FFFF0FFF

Note: As shown in Table 4, "Intel® SS72 NAND Flash Memory Array Addressing (x16)" on page 12, the upper five bits of ADDRESS cycle 2 have no assigned address bits; however, these 5 bits must be held LOW during the ADDRESS cycle to ensure that the address is interpreted correctly by the NAND Flash device. These extra bits are accounted for in ADDRESS cycle 2 even though they do not have address bits assigned to them.

Figure 5. Intel® SS72 NAND Flash Memory Array Organization (x16)

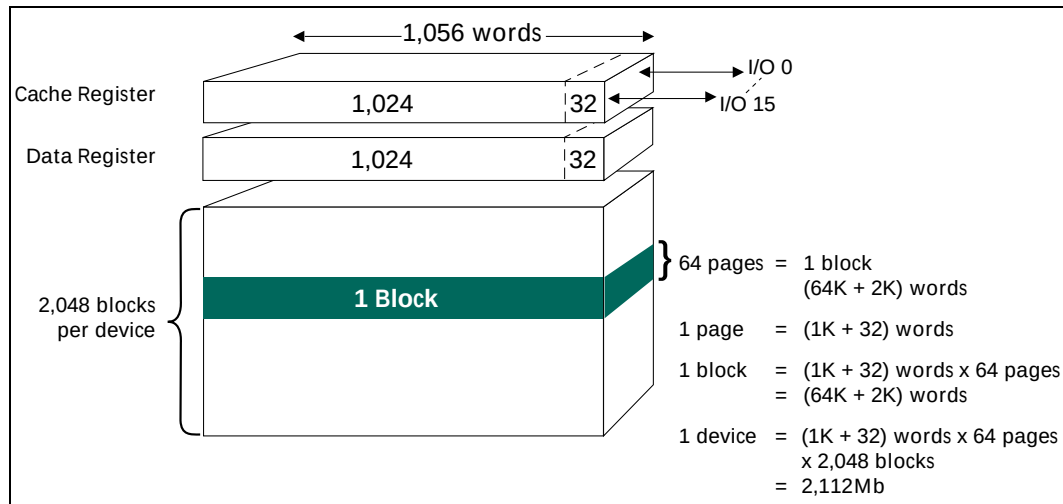


Table 4. Intel® SS72 NAND Flash Memory Array Addressing (x16)

Cycle	I/O[15:8]	I/O7	I/O6	I/O5	I/O4	I/O3	I/O2	I/O1	I/O0
First	LOW	CA7	CA6	CA5	CA4	CA3	CA2	CA1	CA0
Second	LOW	LOW	LOW	LOW	LOW	LOW	CA10 ¹	CA9	CA8
Third	LOW	BA7	BA6	PA5	PA4	PA3	PA2	PA1	PA0
Fourth	LOW	BA15	BA14	BA13	BA12	BA11	BA10	BA9	BA8
Fifth	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW	BA16

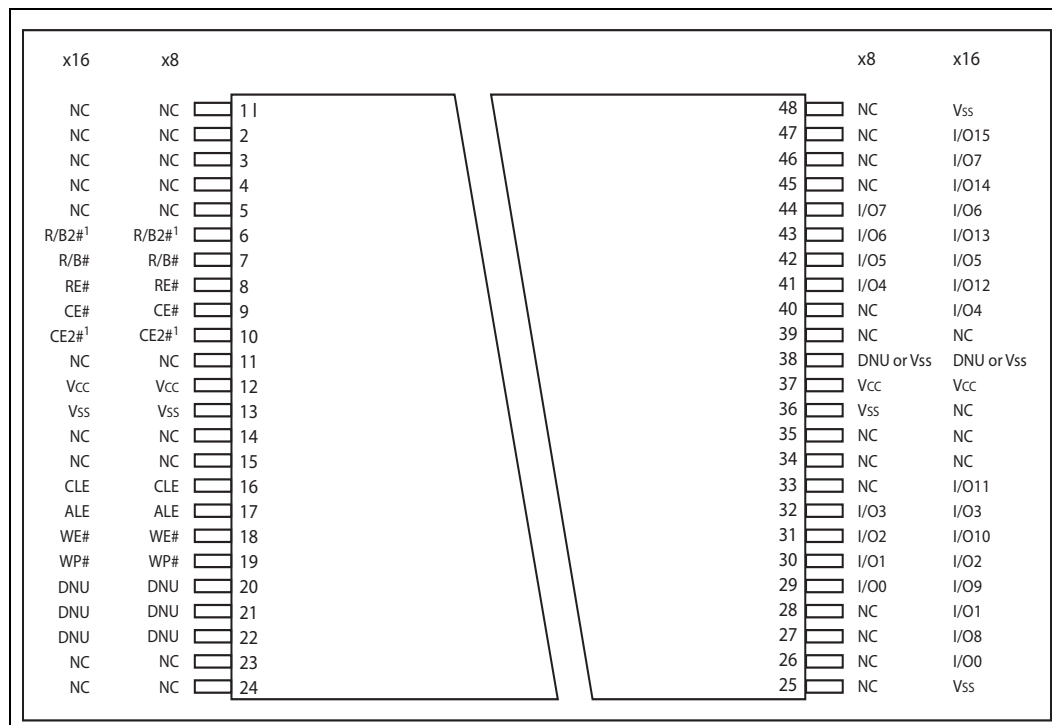
Notes:

1. If CA10 = "1" then CA[9:5] must be "0."
2. Block address concatenated with page address = actual page address. CAx = column address; PAx = page address, BAx = block address.
3. I/O[15:8] are not used during the addressing sequence and should be driven LOW.

§ §

2.0 Signal Assignments and Descriptions

Figure 6. 48-Pin TSOP Type 1 Pin Assignment Top View

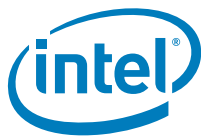


Notes:

1. R/B2# and CE2# are only available on 8Gb devices. These pins are NC for other configurations.

Table 5. Signal Descriptions

Symbol	Type	Description
ALE	Input	Address latch enable: During the time ALE is HIGH, address information is transferred from I/O[7:0] into the on-chip address register upon a LOW-to-HIGH transition on WE#. When address information is not being loaded, the ALE pin should be driven LOW.
CE#, CE2#	Input	Chip enable: This gates transfers between the host system and the NAND Flash device. Once the device starts a PROGRAM or ERASE operation, the chip enable pin can be de-asserted. For the 8Gb configuration, CE# controls the first 4Gb of memory; CE2# controls the second 4Gb. See Section 5.0, "NAND Flash Bus Operations" on page 21 for additional operational details. In the 8Gb configuration, R/B# is for the 4Gb of memory enabled by CE#; R/B2# is for the 4Gb of memory enabled by the CE2#.
CLE	Input	Command latch enable: When CLE is HIGH, information is transferred from I/O[7:0] to the on-chip command register on the rising edge of WE#. When command information is not being loaded, CLE should be driven LOW.
RE#	Input	Read enable: This gates transfers from the NAND Flash device to the host system.
WE#	Input	Write enable: This gates transfers from the host system to the NAND Flash device.
WP#	Input	Write protect: Pin protects against inadvertent PROGRAM and ERASE operations. All PROGRAM and ERASE operations are disabled when the WP# pin is LOW.

**Table 5. Signal Descriptions**

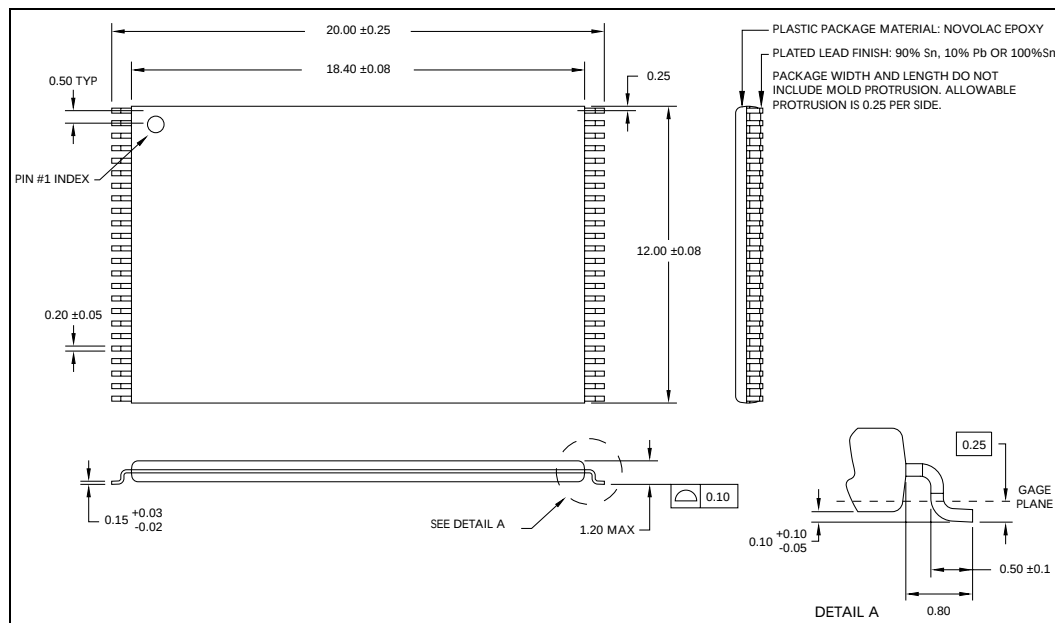
Symbol	Type	Description
I/O[7:0] (8-bit bus width) I/O[15:0] (16-bit bus width)	I/O	Data inputs/outputs: The bidirectional I/O pins transfer address, data, and instruction information. Data is output only during READ operations; at other times the I/O pins are inputs.
R/B#, R/B2#	Output	Ready/busy: An open-drain, active-LOW output that uses an external pull-up resistor, the pin is used to indicate when the chip is processing a PROGRAM or ERASE operation. The pin is also used during a READ operation to indicate when data is being transferred from the array into the serial data register. When these operations have completed, the R/B# returns to the High-Z state.
Vcc	Supply	Vcc: Power supply.
Vss	Supply	Vss: Ground connection.
NC	–	No connect: NC pins are not internally connected. These pins can be driven or left unconnected.
DNU	–	Do not use: These pins must be left unconnected.

§ §



3.0 Package Information

Figure 7. TSOP Type 1 Package



Note: All dimensions in millimeters, MIN/MAX, or typical, as noted.

§ §

4.0 Electrical Characteristics

4.1 Maximum Ratings and Operating Conditions

Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating *only*, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not guaranteed. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

Table 6. Absolute Maximum Ratings by Device: Voltage on any pin relative to VSS

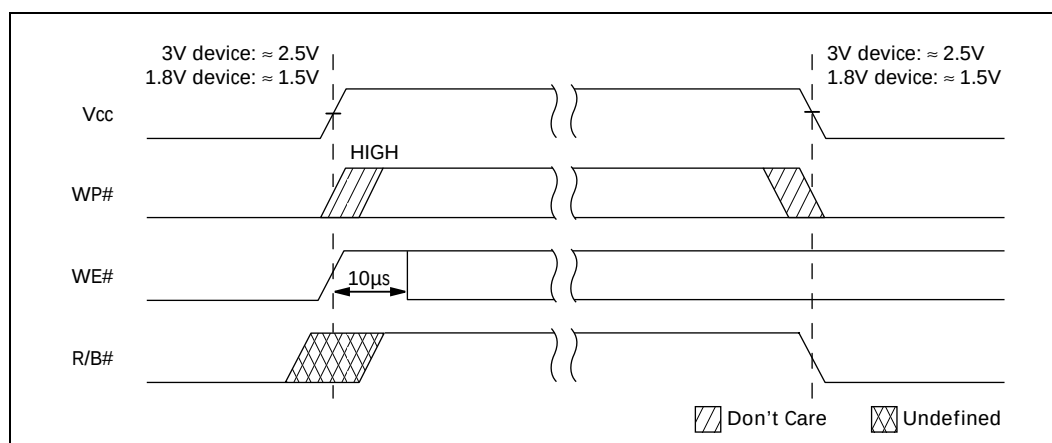
Parameter/Condition		Symbol	Min	Max	Unit
Voltage input	JS29F02G08AANB3	VIN	-0.6	+4.6	V
	JS29F04G08BANB3				
	JS29F08G08FANB3				
Voltage input	(1.8V version)	VIN	-0.6	+2.4	V
Vcc supply voltage	JS29F02G08AANB3	Vcc	-0.6	+4.6	V
	JS29F04G08BANB3				
	JS29F08G08FANB3				
Vcc supply voltage	(1.8V version)	VIN	-0.6	+2.4	V
Storage temperature		TSTG	-65	+150	°C
Short circuit output current, I/Os			—	5	mA

Table 7. Recommended Operating Conditions

Parameter/Condition		Symbol	Min	Typ	Max	Unit
Operating temperature	Commercial	TA	0	—	+70	°C
	Extended		-40	—	+85	°C
Vcc supply voltage	JS29F02G08AANB3	Vcc	2.7	3.3	3.6	V
	JS29F04G08BANB3					
	JS29F08G08FANB3					
Vcc supply voltage	(1.8V version)	Vcc	1.70	1.8	1.95	V
Ground supply voltage		Vss	0	0	0	V

4.2 Vcc Power Cycling

Intel® NAND Flash devices are designed to prevent data corruption during power transitions. Vcc is internally monitored. When Vcc goes below approximately 1.1V (1.8V device), or 2.0V (3V device), PROGRAM and ERASE functions are disabled. WP# provides additional hardware protection. WP# should be kept at VIL during power cycling. When Vcc reaches 1.5V (1.8V device) or 2.5V (3V device), a minimum of 10μs should be allowed for the Flash to initialize before executing any commands.

**Figure 8. AC Waveforms During Power Transitions****Table 8. DC and Operating Characteristics**

Parameter		Conditions	Symbol	Min	Typ	Max	Unit
Sequential read current		$t_{RC} = 30ns$, $CE\# = V_{IL}$, $I_{OUT} = 0mA$	Icc1	–	15	30	mA
Program current		–	Icc2	–	15	30	mA
Erase current		–	Icc3	–	15	30	mA
Standby current (TTL)		$CE\# = V_{IH}$, $WP\# = 0V/V_{CC}$	ISB1	–	–	1	mA
Standby current (CMOS)							
	JS29F02G08AANB3	$CE\# = V_{CC} - 0.2V$, $WP\# = 0V/V_{CC}$	ISB2	–	10	50	μA
	JS29F04G08BANB3			–	20	100	μA
	JS29F08G08FANB3			–	40	200	μA
Input leakage current							
	JS29F02G08AANB3	$V_{IN} = 0V \text{ to } V_{CC}$	ILI	–	–	±10	μA
	JS29F04G08BANB3			–	–	±20	μA
	JS29F08G08FANB3			–	–	±40	μA
Output leakage current							
	JS29F02G08AANB3	$V_{OUT} = 0V \text{ to } V_{CC}$	ILO	–	–	±10	μA
	JS29F04G08BANB3			–	–	±20	μA
	JS29F08G08FANB3			–	–	±40	μA
Input high voltage		I/O[7:0], I/O[15:0] CE#, CLE, ALE, WE#, RE#, WP#, R/B#	V _{IH}	0.8 x V _{CC}	–	V _{CC} + 0.3	V
Input low voltage (all inputs)		–	V _{IL}	–0.3	–	0.2 x V _{CC}	V
Output high voltage		I _{OH} = –400μA	V _{OH}	2.4	–	–	V
Output low voltage		I _{OL} = 2.1mA	V _{OL}	–	–	0.4	V
Output low current (R/B#)		V _{OL} = 0.4V	I _{OL} (R/B#)	8	10	–	mA

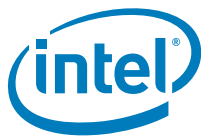


Table 9. Valid Blocks

Parameter	Symbol	Device	Min	Max	Unit	Notes
Valid block number	NVB	JS29F02G08AANB3	2,008	2,048	Blocks	1, 2
		JS29F04G08BANB3	4,016	4,096		
		JS29F08G08FANB3	8,032	8,192		3

Notes:

- Invalid blocks are blocks that contain one or more bad bits. The device may contain bad blocks upon shipment. Additional bad blocks may develop over time; however, the total number of available blocks will not drop below NVB during the endurance life of the device. Do not erase or program blocks marked invalid by the factory.
- Block 00h (the first block) is guaranteed to be valid and does not require error correction up to 1K PROGRAM/ERASE cycles.
- The number of invalid blocks in each 4Gb section will not exceed 80.

Table 10. Capacitance

Description	Symbol	Device	Max	Unit	Notes
Input capacitance	C _{IN}	JS29F02G08AANB3	10	pF	1, 2
		JS29F04G08BANB3	20		
		JS29F08G08FANB3	40		
Input/output capacitance (I/O)	C _{IO}	JS29F02G08AANB3	10	pF	1, 2
		JS29F04G08BANB3	20		
		JS29F08G08FANB3	40		

Notes:

- These parameters are verified in device characterization and are not 100 percent tested.
- Test conditions: T_c = 25°C; f = 1 MHz; V_{IN} = 0V.

Table 11. Test Conditions

Parameter		Value	Notes
Input pulse level	JS29F02G08AANB3	0.0V to V _{CC} (2.7V–3.6V)	
	JS29F04G08BANB3		
	JS29F08G08FANB3		
Input pulse level	(1.8V version)	0.0V to V _{CC} (1.70V–1.95V)	
Input rise and fall times		5ns	
Input and output timing levels		V _{CC} /2	
Output load	V _{CC} = 3.0V ± 10%	1 TTL GATE and CL = 50pF	1
	V _{CC} = 3.3V ± 10%	1 TTL GATE and CL = 100pF	1
Output load	V _{CC} = 1.70V–1.95V	1 TTL GATE and CL = 30pF	1

Notes:

- Verified in device characterization; not 100 percent tested.

**Table 12. AC Characteristics: Command, Data, and Address Input**

Parameter	Symbol	3V x16		3V x8		Unit	Notes
		Min	Max	Min	Max		
ALE to data start	t_{ADL}	100	—	100	—	ns	1
ALE hold time	t_{ALH}	10	—	5	—	ns	2
ALE setup time	t_{ALS}	25	—	10	—	ns	2
CE# hold time	t_{CH}	10	—	5	—	ns	2
CLE hold time	t_{CLH}	10	—	5	—	ns	2
CLE setup time	t_{CLS}	25	—	10	—	ns	2
CE# setup time	t_{CS}	35	—	15	—	ns	2
Data hold time	t_{DH}	10	—	5	—	ns	2
Data setup time	t_{DS}	20	—	10	—	ns	2
Write cycle time	t_{WC}	45	—	30	—	ns	2
WE# pulse width HIGH	t_{WH}	15	—	10	—	ns	2
WE# pulse width	t_{WP}	25	—	15	—	ns	2
WP# setup time	t_{WW}	30	—	30	—	ns	

Notes:

- Timing for t_{ADL} begins in the ADDRESS cycle, on the final rising edge of WE#, and ends with the first rising edge of WE# for data input.
- For PAGE READ CACHE MODE and PROGRAM PAGE CACHE MODE operations, the 3V x16 AC characteristics apply for 3V x8 devices.

Table 13. AC Characteristics: Normal Operation (Sheet 1 of 2)

Parameter	Symbol	3V x16		3V x8		Unit	Notes
		Min	Max	Min	Max		
ALE to RE# delay	t_{AR}	10	—	10	—	ns	
CE# access time	t_{CEA}	—	45	—	23	ns	1
CE# HIGH to output High-Z	t_{CHZ}	—	20	—	20	ns	2
CLE to RE# delay	t_{CLR}	10	—	10	—	ns	
Cache busy in page read cache mode (first 31h)	$t_{DCBSYR1}$	—	3	—	3	μs	
Cache busy in page read cache mode (next 31h and 3Fh)	$t_{DCBSYR2}$	$t_{DCBSYR1}$	25	$t_{DCBSYR1}$	25	μs	
Output High-Z to RE# LOW	t_{IR}	0	—	0	—	ns	1
Data output hold time	t_{OH}	15	—	15	—	ns	
Data transfer from NAND Flash array to data register	t_R	—	25	—	25	μs	
READ cycle time	t_{RC}	50	—	30	—	ns	1
RE# access time	t_{REA}	—	30	—	18	ns	1
RE# HIGH hold time	t_{REH}	15	—	10	—	ns	1
RE# HIGH to output High-Z	t_{RHZ}	—	30	—	30	ns	2
RE# pulse width	t_{RP}	25	—	15	—	ns	1
Ready to RE# LOW	t_{RR}	20	—	20	—	ns	

**Table 13. AC Characteristics: Normal Operation (Sheet 2 of 2)**

Parameter	Symbol	3V x16		3V x8		Unit	Notes
		Min	Max	Min	Max		
Reset time (READ/ PROGRAM/ERASE)	t_{RST}	–	5/10/ 500	–	5/10/ 500	μs	3
WE# HIGH to busy	t_{WB}	–	100	–	100	ns	3, 4
WE# HIGH to RE# LOW	t_{WHR}	60	–	60	–	ns	

Notes:

1. For PAGE READ CACHE MODE and PROGRAM PAGE CACHE MODE operations, the 3V x16 AC characteristics apply for 3V x8 devices.
2. Transition is measured $\pm 200mV$ from steady-state voltage with load. This parameter is sampled and not 100 percent tested.
3. If RESET (FFh) command is loaded at ready state, the device goes busy for maximum 5 μs .
4. Do not issue a new command during t_{WB} , even if R/B# is ready.

Table 14. PROGRAM/ERASE Characteristics

Parameter	Description	Typ	Max	Unit	Notes
NOP	Number of partial page programs	—	8	Cycles	1
t_{BERS}	Block erase time	2	3	ms	
t_{CBSY}	Busy time for cache program	3	700	μs	2
t_{LPROG}	Last page program time	—	—	—	3
t_{PROG}	Page program time	300	700	μs	

Notes:

1. Eight total to the same page.
2. t_{CBSY} MAX time depends on timing between internal program completion and data in.
3. $t_{LPROG} = t_{PROG}(\text{last page}) + t_{PROG}(\text{last} - 1 \text{ page}) - \text{command load time (last page)} - \text{address load time (last page)} - \text{data load time (last page)}$.

§ §



5.0 NAND Flash Bus Operations

The bus on the 8-bit bus width Intel® SS72 NAND Flash Memory device is multiplexed. Data I/O, addresses, and commands all share the same pins. I/O pins I/O[15:8] are used only for data in the x16 configuration. Addresses and commands are always supplied on I/O[7:0].

The command sequence normally consists of a command latch cycle, an ADDRESS LATCH cycle, and a DATA cycle—either READ or WRITE.

5.1 Control Signals

CE#, WE#, RE#, CLE, ALE and WP# control NAND Flash device READ and WRITE operations. On the 8Gb JS29F08G08FANB3, CE# and CE2# each control independent 4Gb arrays. CE2# functions the same as CE# for its own array; all operation described for CE# also apply to CE2#.

CE# is used to enable the device. When CE# is LOW and the device is not in the busy state, the NAND Flash memory will accept command, data, and address information.

When the device is not performing an operation, the CE# pin is typically driven HIGH and the device enters standby mode. The memory will enter standby if CE# goes HIGH while data is being transferred and the device is not busy. This helps reduce power consumption. See [Figure 39, "PAGE READ Operation with CE# Don't Care" on page 47](#) and [Figure 47, "PROGRAM PAGE Operation with CE# Don't Care" on page 51](#) for examples of CE# "Don't Care" operations.

The CE# "Don't Care" operation enables the NAND Flash to reside on the same asynchronous memory bus as other Flash or SRAM devices. Other devices on the memory bus can then be accessed while the NAND Flash is busy with internal operations. This capability is important for designs that require multiple NAND devices on the same bus. One device can be programmed while another is being read.

A HIGH CLE signal indicates that a command cycle is taking place. A HIGH ALE signal signifies that an address input cycle is occurring.

5.2 Commands

Commands are written to the command register on the rising edge of WE# when:

- CE# and ALE are LOW, and
- CLE is HIGH, and
- the device is not busy.

The exceptions to this are the READ STATUS and RESET commands when busy. See [Figure 34, "ADDRESS LATCH Cycle" on page 44](#) for detailed timing requirements.

Commands are input on I/O[7:0] only. For devices with a x16 interface, I/O[15:8] must be written with zeros when issuing a command.

5.3 Address Input

Addresses are written to the address register on the rising edge of WE# when:

- CE# and CLE are LOW, and
- ALE is HIGH, and
- the device is not busy.

Addresses are input on I/O[7:0] only; bits not part of the address space must be LOW. For devices with a x16 interface, I/O[15:8] must be written with zeros when issuing an address.

The number of ADDRESS cycles required for each command varies. Refer to the command descriptions to determine addressing requirements ([Table 16, "Command Set" on page 26](#)).

5.4 Data Input

Data is written to the data register on the rising edge of WE# when:

- CE#, CLE, and ALE are LOW, and
- the device is not busy.

Data is input on I/O[7:0] for x8 devices, and I/O[15:0] on x16 devices. See [Figure 35, "INPUT DATA LATCH" on page 45](#) for additional data input details.

5.5 READs

After a READ command is issued, data is transferred from the memory array to the data register from the rising edge of WE#. R/B# goes LOW for t_R and transitions HIGH after the transfer is complete. When R/B# goes HIGH, data is available in the data register, and is clocked out of the part by toggling RE#. See [Figure 38, "PAGE READ" on page 46](#) for detailed timing information.

The READ STATUS (70h) command or the R/B# signal can be used to determine when the device is ready. See [Section 6.1.5, "READ STATUS 70h" on page 30](#) for details.

5.6 Ready/Busy#

The R/B# output provides a hardware method of indicating the completion of PROGRAM, ERASE, and READ operations. The signal requires a pull-up resistor for proper operation. The signal is typically HIGH, and transitions to LOW after the appropriate command is written to the device. The signal pin's open-drain driver enables multiple R/B# outputs to be OR-tied. The READ STATUS command can be used in place of R/B#. Typically R/B# is connected to an interrupt pin on the system controller (see [Figure 10, "READY/BUSY# Open Drain" on page 23](#)).

On the 8Gb JS29F08G08FANB3, R/B# provides a status indication for the 4Gb section enabled by CE#, and R/B2# does the same for the 4Gb section enabled by CE2#. R/B# and R/B2# can be tied together, or they can be used separately to provide independent indications for each 4Gb section.

The combination of R_p and capacitive loading of the R/B# circuit determines the rise time of the R/B# pin. The actual value used for R_p depends on the system timing requirements. Large values of R_p cause R/B# to be delayed significantly. At the 10 to 90 percent points on the R/B# waveform, rise time is approximately two time constants (TC).

$$TC = R * C$$

where $R = R_p$ (resistance of pull-up resistor), and C = total capacitive load

The fall time of the R/B# signal is determined mainly by the output impedance of the R/B# pin and the total load capacitance.



Refer to Figure 11, " t_{Fall} and t_{Rise} " on page 24, and Figure 12, " I_{OL} vs. R_p " on page 24, which depict approximate R_p values using a circuit load of 100pF.

The minimum value for R_p is determined by the output drive capability of the R/B# signal, the output voltage swing, and V_{CC} .

Figure 9. Minimum R_p

$$R_p (\text{MIN, 1.8V part}) = \frac{V_{\text{CC}} (\text{MAX}) - V_{\text{OL}} (\text{MAX})}{I_{\text{OL}} + \Sigma I_L} = \frac{1.85\text{V}}{3\text{mA} + \Sigma I_L}$$

$$R_p (\text{MIN, 3.3V part}) = \frac{V_{\text{CC}} (\text{MAX}) - V_{\text{OL}} (\text{MAX})}{I_{\text{OL}} + \Sigma I_L} = \frac{3.2\text{V}}{8\text{mA} + \Sigma I_L}$$

Where ΣI_L is the sum of the input currents of all devices tied to the R/B# pin.

Figure 10. READY/BUSY# Open Drain

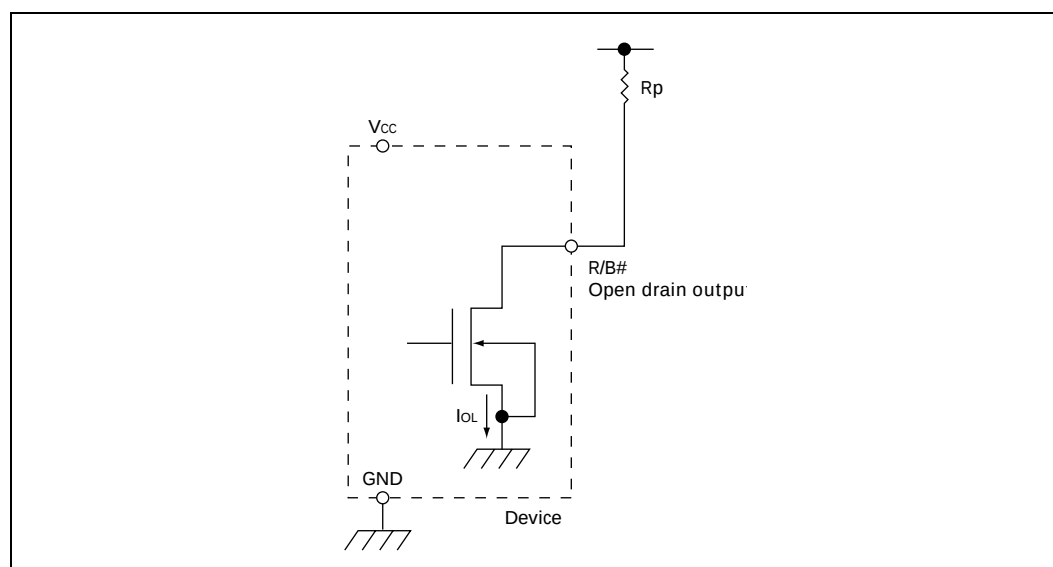
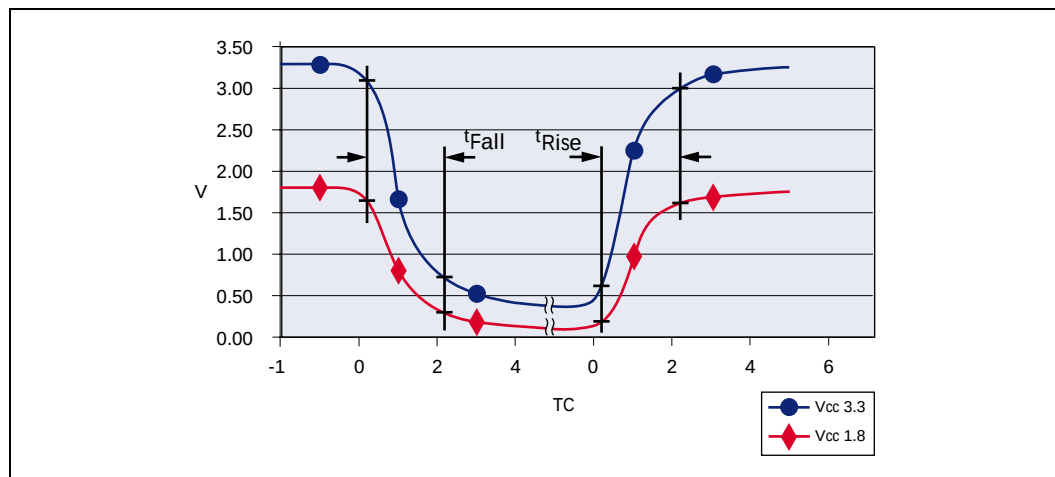


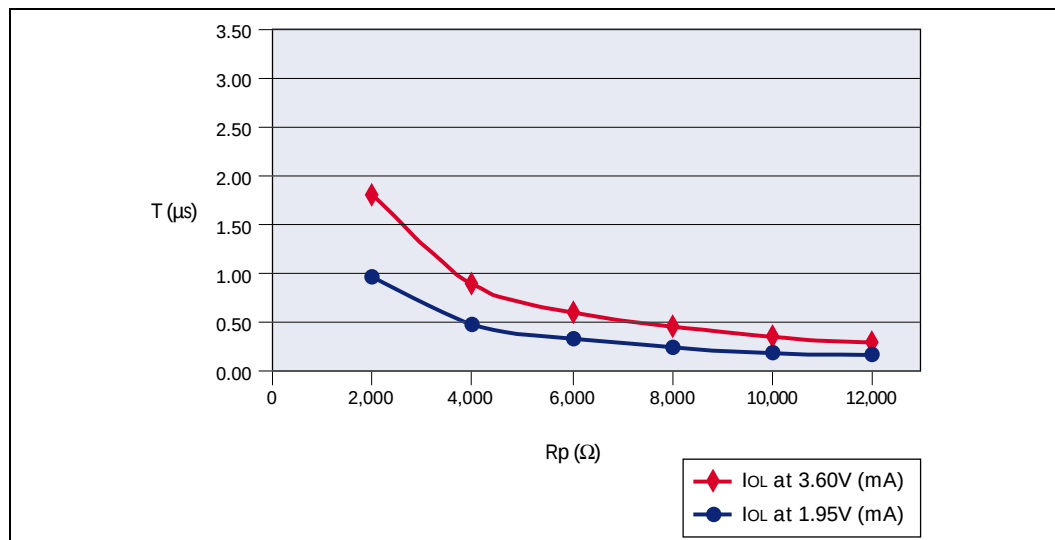
Figure 11. t_{Fall} and t_{Rise}



Notes:

1. t_{Fall} and t_{Rise} calculated at 10 percent–90 percent points.
2. t_{Rise} dependent on external capacitance, and resistive loading and output-transistor impedance.
3. t_{Rise} primarily dependent on external pull-up resistor and external capacitive loading.
4. $t_{Fall} \approx 10\text{ns}$ at 3.3V; $t_{Fall} \approx 7\text{ns}$ at 1.8V.
5. See TC values in Figure 12, “ I_{OL} vs. R_p ” on page 24 for approximate R_p value and TC.

Figure 12. I_{OL} vs. R_p



**Table 15. Mode Selection**

CLE	ALE	CE#	WE#	RE#	WP# ¹	Mode	
H	L	L		H	X	Read mode	Command input
L	H	L		H	X		Address input
H	L	L		H	H	Write mode	Command input
L	H	L		H	H		Address input
L	L	L		H	H	Data input	
L	L	L	H		X	Sequential read and data output	
L	L	L	H	H	X	During read (busy)	
X	X	X	X	X	H	During program (busy)	
X	X	X	X	X	H	During erase (busy)	
X	X	X	X	X	L	Write protect	
X	X	H	X	X	0V/Vcc	Standby	

Notes:

1. WP# should be biased to CMOS HIGH or LOW for standby.
2. Mode selection settings for this table: H = Logic level HIGH; L = Logic level LOW; X = V_{IH} or V_{IL}.





6.0 Command Definitions

Table 16. Command Set

Operation	Command Cycle 1	Number of Address Cycles	Data Cycles Required ¹	Command Cycle 2	Valid During Busy	Notes
PAGE READ	00h	5	No	30h	No	
PAGE READ CACHE MODE START	31h	—	No	—	No	
PAGE READ CACHE MODE START LAST	3Fh	—	No	—	No	
READ for INTERNAL DATA MOVE	00h	5	No	35h	No	2
RANDOM DATA READ	05h	2	No	E0h	No	3
READ ID	90h	1	No	—	No	
READ STATUS	70h	—	No	—	Yes	
PROGRAM PAGE	80h	5	Yes	10h	No	
PROGRAM PAGE CACHE MODE	80h	5	Yes	15h	No	
PROGRAM for INTERNAL DATA MOVE	85h	5	Optional	10h	No	2
RANDOM DATA INPUT	85h	2	Yes	—	No	4
BLOCK ERASE	60h	3	No	D0h	No	
RESET	FFh	—	No	—	Yes	
OTP DATA PROGRAM	A0h	5	Yes	10h	No	
OTP DATA PROTECT	A5h	5	No	10h	No	
OTP DATA READ	AFh	5	No	30h	No	

Notes:

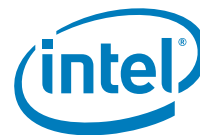
1. Indicates required data cycles between command cycle 1 and command cycle 2.
2. Do not cross die boundaries when using READ for INTERNAL DATA MOVE and PROGRAM for INTERNAL DATA MOVE.
3. RANDOM DATA READ command limited to use within a single page.
4. RANDOM DATA INPUT command limited to use within a single page.

6.1 READ Operations

6.1.1 PAGE READ 00h-30h

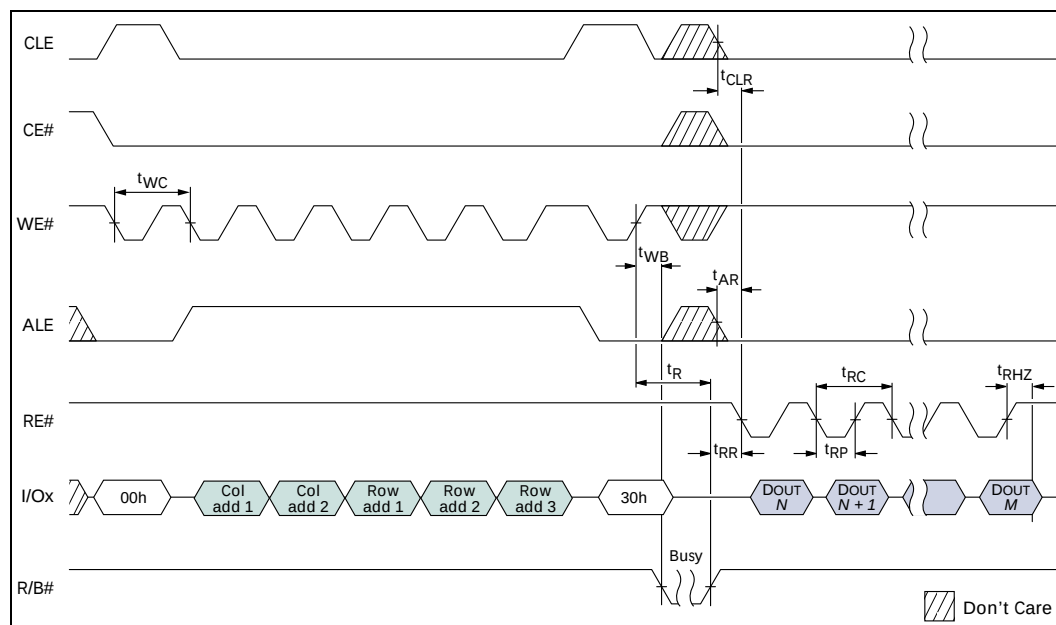
On initial power up, the device defaults to read mode. To enter the read mode while in operation, write the 00h command to the command register, then write five ADDRESS cycles followed by the 30h command.

To determine the progress of the data transfer from the NAND Flash array to the data register (^tR), monitor the R/B# signal; or alternately, issue a READ STATUS (70h) command. If the READ STATUS command is used to monitor the data transfer, the user must re-issue the READ (00h) command to receive data output from the data register. See [Figure 43, "PAGE READ CACHE MODE Timing without R/B# \(Part 1 of 2\)" on page 49](#) and [Figure 44, "PAGE READ CACHE MODE Timing without R/B# \(Part 2 of 2\)" on page 49](#) for examples. After the READ command has been re-issued, pulsing the RE# line will result in outputting data, starting from the initial column address.



A serial page read sequence outputs a complete page of data. After 30h is written, the page data is transferred to the data register, and R/B# goes LOW during the transfer. When the transfer to the data register is complete, R/B# returns HIGH. At this point, data can be read from the device. Starting from the initial column address to the end of the page, read the data by repeatedly pulsing RE# at the maximum t_{RC} rate.

Figure 13. PAGE READ Operation



6.1.2 RANDOM DATA READ 05h-E0h

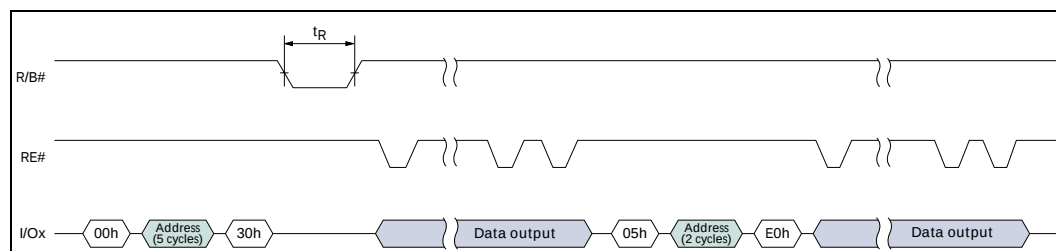
The RANDOM DATA READ command enables the user to specify a new column address so the data at single or multiple addresses can be read. The random read mode is enabled after a normal PAGE READ (00h-30h) sequence.

Random data can be output after the initial page read by writing an 05h-E0h command sequence along with the new column address (two cycles).

The RANDOM DATA READ command can be issued without limit within the page.

Only data on the current page can be read. Pulsing the RE# pin outputs data sequentially.

Figure 14. RANDOM DATA READ Operation



6.1.3 PAGE READ CACHE MODE START 31h; PAGE READ CACHE MODE START LAST 3Fh

Intel® NAND Flash devices have a cache register that can be used to increase the READ operation speed when accessing sequential pages in a block.

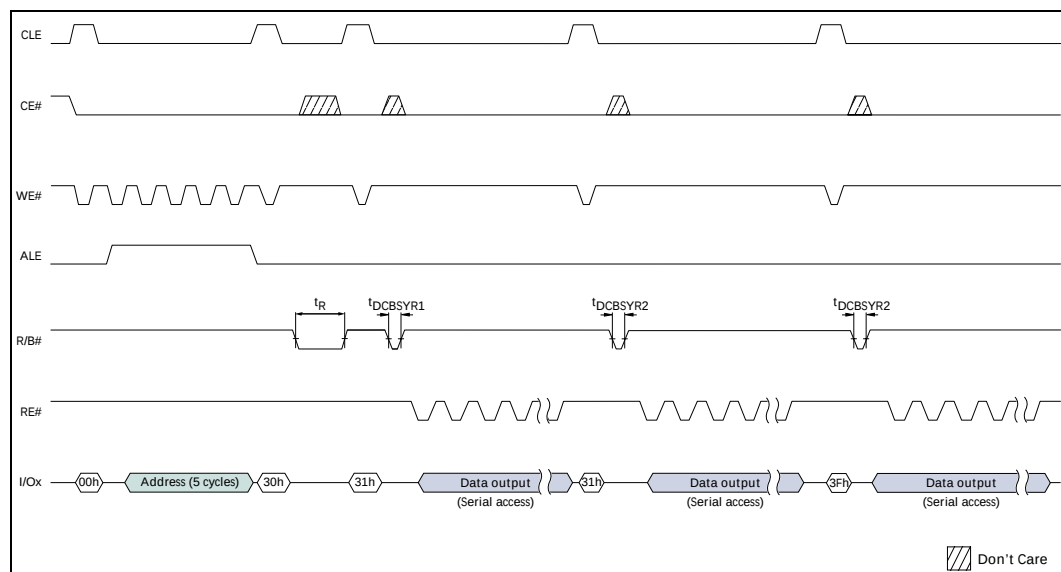
First, a normal PAGE READ (00h-30h) command sequence is issued. The R/B# signal goes LOW for t_R during the time it takes to transfer the first page of data from the memory to the data register. After R/B# returns to HIGH, the PAGE READ CACHE MODE START (31h) command is latched into the command register. R/B# goes LOW for $t_{DCBSYR1}$ while data is being transferred from the data register to the cache register. Once the data register contents are transferred to the cache register, another PAGE READ is automatically started as part of the 31h command. Data is transferred from the next sequential page of the memory array to the data register during the same time data is being read serially (pulsing of RE#) from the cache register. If the total time to output data exceeds t_R , then the PAGE READ is hidden.

The second and subsequent pages of data are transferred to the cache register by issuing additional 31h commands. R/B# will stay LOW up to $t_{DCBSYR2}$. This time can vary, depending on whether the previous memory-to-data-register transfer was completed prior to issuing the next 31h command. See [Table 13, "AC Characteristics: Normal Operation"](#) on page 19 for timing parameters. If the data transfer from memory to the data register is not completed before the 31h command is issued, R/B# stays LOW until the transfer is complete.

It is not necessary to output a whole page of data before issuing another 31h command. R/B# will stay LOW until the previous PAGE READ is complete and the data has been transferred to the cache register.

To read out the last page of data, the PAGE READ CACHE MODE START LAST (3Fh) command is issued. This command transfers data from the data register to the cache register without issuing another PAGE READ.

Figure 15. PAGE READ CACHE MODE



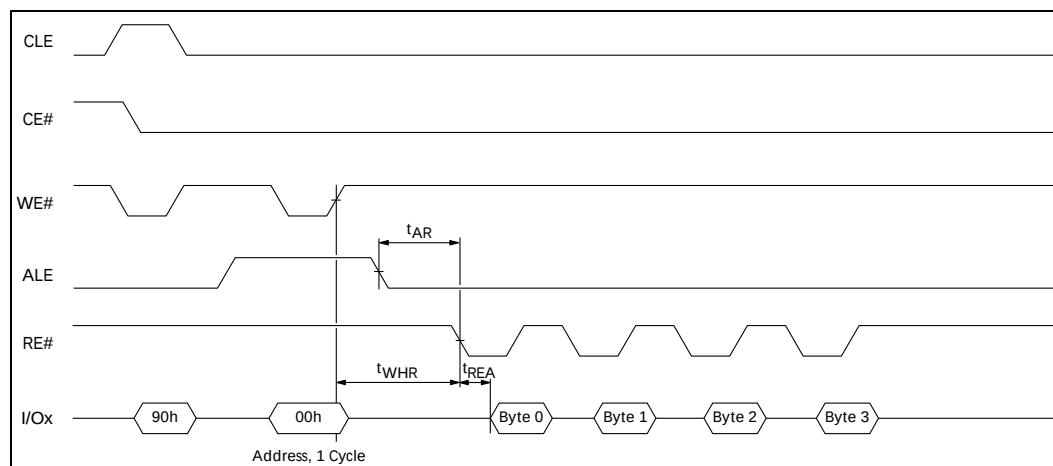


6.1.4 READ ID 90h

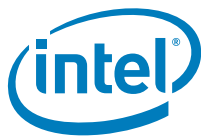
The READ ID command is used to read the 4 bytes of identifier codes programmed into the devices. The READ ID command reads a 4-byte table that includes Manufacturer ID, device configuration, and part-specific information (see [Table 17, "Device ID and Configuration Codes" on page 30](#)).

Writing 90h to the command register puts the device into the read ID mode. The command register stays in this mode until another valid command is issued (see [Figure 16, "READ ID Operation" on page 29](#)).

Figure 16. READ ID Operation



Note: See [Table 17, "Device ID and Configuration Codes" on page 30](#) for byte definitions.


Table 17. Device ID and Configuration Codes

	Options	I/O 7	I/O 6	I/O 5	I/O 4	I/O 3	I/O 2	I/O 1	I/O 0	Value ¹	Notes
Byte 0	Manufacturer ID										
	Intel®	0	0	1	0	1	1	0	0	2Ch	
Byte 1	Device ID										
JS29F02G08AANB3	2Gb, x8, 3V	1	1	0	1	1	0	1	0	DAh	
JS29F04G08BANB3	4Gb, x8, 3V	1	1	0	1	1	1	0	0	DCh	
JS29F08G08FANB3	8Gb, x8, 3V	1	1	0	1	1	1	0	0	DCh	3
Byte 2											
Byte value	Don't Care	x	x	x	x	x	x	x	x	XXh	
Byte 3											
Page size	2KB							0	1	01b	
Spare area size (bytes)	64					0	1			01b	
Block size (w/o spare)	128KB			0	1					01b	
Organization	x8		0							0b	
Reserved		0								0b	
Byte value	x8	0	0	0	1	0	1	0	1	15h	

Notes:

1. b = binary, h = hex.
2. Device IDs for these configurations are provided for reference only.
3. The JS29F08G08FANB3 device ID code reflects the configuration of each 4Gb section.

6.1.5 READ STATUS 70h

These NAND Flash devices have an 8-bit status register that the software can read during device operation. On the x16 device, I/O[15:8] are "0" when reading the status register. [Table 18](#) describes the status register.

After a READ STATUS command, all READ cycles will be from the status register until a new command is issued. Changes in the status register will be seen on I/O[7:0] as long as CE# and RE# are LOW; it is not necessary to start a new READ STATUS cycle to see these changes.

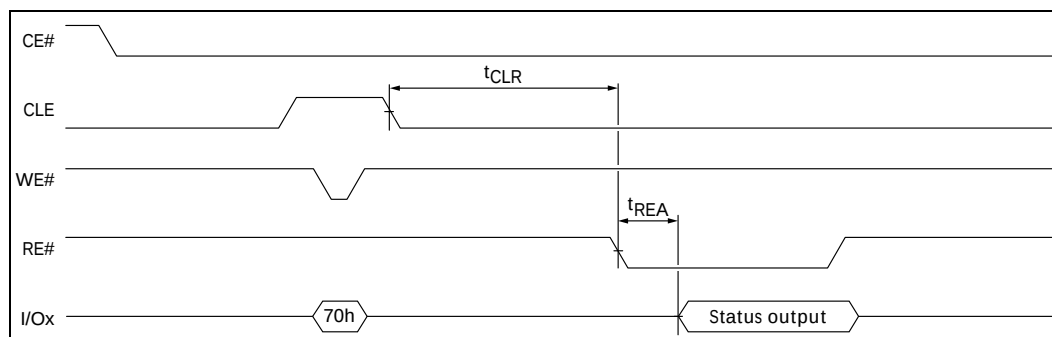
While monitoring the read status to determine when the t_R (transfer from NAND Flash array to data register) is complete, the user must re-issue the READ (00h) command to make the change from status mode to data mode. After the READ command has been re-issued, pulsing the RE# line will result in outputting data, starting from the initial column address.

**Table 18. Status Register Bit Definition**

SR Bit	Page Program	Program Page Cache Mode	Page Read	Page Read Cache Mode	Block Erase	Definition
0	Pass/fail	Pass/fail (N)	—	—	Pass/fail	"0" = Successful PROGRAM/ERASE "1" = Error in PROGRAM/ERASE
1	—	Pass/fail (N-1)	—	—	—	"0" = Successful PROGRAM/ERASE "1" = Error in PROGRAM/ERASE
2	—	—	—	—	—	"0"
3	—	—	—	—	—	"0"
4	—	—	—	—	—	"0"
5	Ready/busy	Ready/busy ¹	Ready/busy	Ready/busy ¹	Ready/busy	"0" = Busy "1" = Ready
6	Ready/busy	Ready/busy cache ²	Ready/busy	Ready/busy cache ²	Ready/busy	"0" = Busy "1" = Ready
7	Write protect	Write protect	Write protect	Write protect	Write protect	"0" = Protected "1" = Not protected
[15:8]	—	—	—	—	—	"0"

Notes:

1. Status register bit 5 is "0" during the actual programming operation. If cache mode is used, this bit will be "1" when all internal operations are complete.
2. Status register bit 6 is "1" when the cache is ready to accept new data. R/B# follows bit 6. See [Figure 15, "PAGE READ CACHE MODE" on page 28](#), and [Figure 20, "PROGRAM PAGE CACHE MODE Example" on page 33](#).

Figure 17. Status Register Operation

6.2 PROGRAM Operations

6.2.1 PROGRAM PAGE 80h-10h

Intel® NAND Flash devices are inherently page-programmed devices. Pages must be programmed consecutively within a block from the least significant page address to the most significant page address (for instance, 0, 1, 2, ..., 63). Random page address programming is prohibited.

Intel® NAND flash devices also support partial-page programming operations. This means that any single bit can only be programmed one time before an erase is required; however, the page can be partitioned such that a maximum of eight programming operations are allowed before an erase is required.

6.2.2 SERIAL DATA INPUT 80h

PROGRAM PAGE operations require loading the SERIAL DATA INPUT (80h) command into the command register, followed by five ADDRESS cycles, then the data. Serial data is loaded on consecutive WE# cycles starting at the given address. The PROGRAM (10h) command is written after the data input is complete. The control logic automatically executes the proper algorithm and controls all the necessary timing to program and verify the operation. Write verification only detects "1s" that are not successfully written to "0s."

R/B# goes LOW for the duration of array programming time, t_{PROG} . The READ STATUS (70h) **command** and the RESET (FFh) command are the only commands valid during the programming operation. Bit 6 of the status register will reflect the state of R/B#. When the device reaches ready, read bit 0 of the status register to determine if the program operation passed or failed (see Figure 18, "PROGRAM and READ STATUS Operation" on page 32). The command register stays in read status register mode until another valid command is written to it.

6.2.3 RANDOM DATA INPUT 85h

After the initial data set is input, additional data can be written to a new column address with the RANDOM DATA INPUT (85h) command. The RANDOM DATA INPUT command can be used any number of times in the same page prior to issuing the PAGE WRITE (10h) command. See Figure 19, "RANDOM DATA INPUT" on page 32 for the proper command sequence.

Figure 18. PROGRAM and READ STATUS Operation

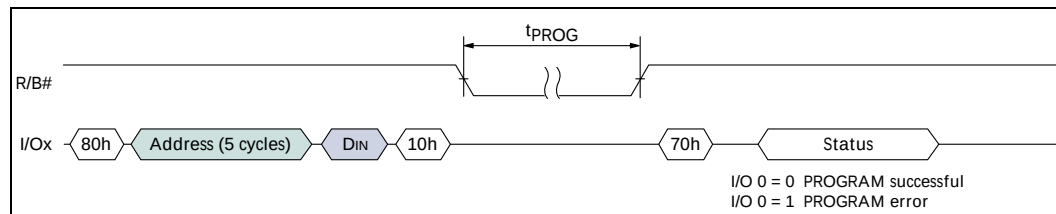
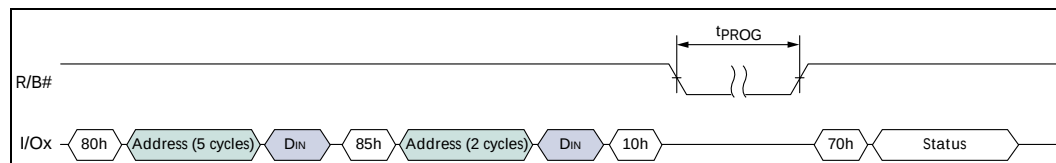


Figure 19. RANDOM DATA INPUT



6.2.4 PROGRAM PAGE CACHE MODE 80h-15h

Cache programming is actually a buffered programming mode of the standard PROGRAM PAGE command. Programming is started by loading the SERIAL DATA INPUT (80h) command to the command register, followed by five cycles of address, and a full or partial page of data. The data is initially copied into the cache register, and the CACHE WRITE (15h) command is then latched to the command register. Data is



transferred from the cache register to the data register on the rising edge of WE#. R/B# goes LOW during this transfer time. After the data has been copied into the data register and R/B# returns to HIGH, memory array programming begins.

When R/B# returns to HIGH, new data can be written to the cache register by issuing another CACHE PROGRAM command sequence. The time that R/B# stays LOW will be controlled by the actual programming time. The first time through equals the time it takes to transfer the cache register contents to the data register. On the second and subsequent programming passes, transfer from the cache register to the data register is held off until current data register content has been programmed into the array.

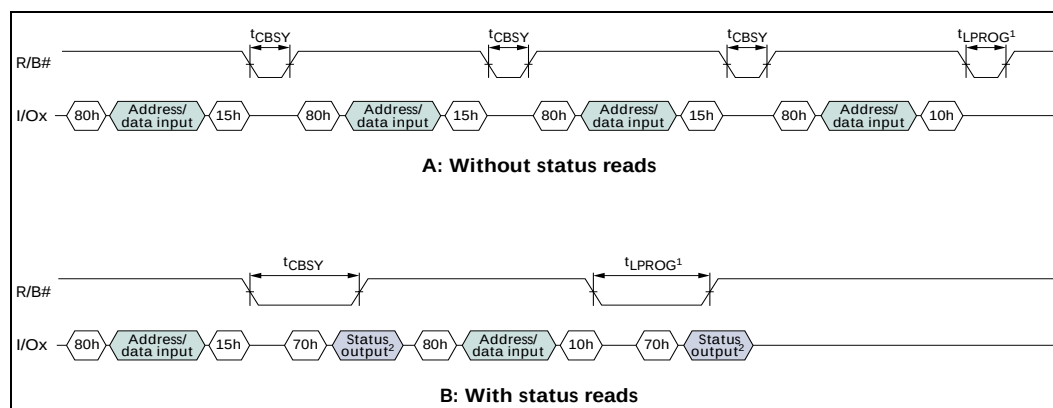
Bit 6 (Cache R/B#) of the status register can be read by issuing the READ STATUS (70h) command to determine when the cache register is ready to accept new data. The R/B# pin always follows bit 6.

Bit 5 (R/B#) of the status register can be polled to determine when the actual programming of the array is complete for the current programming cycle.

If just the R/B# pin is used to determine programming completion, the last page of the program sequence must use the PROGRAM PAGE (10h) command instead of the CACHE PROGRAM (15h) command. If the CACHE PROGRAM (15h) command is used every time, including the last page of the programming sequence, status register bit 5 must be used to determine when programming is complete.

Bit 0 of the status register returns the pass/fail for the previous page when bit 6 of the status register is a "1" (ready state). The pass/fail status of the current PROGRAM operation is returned with bit 0 of the status register when bit 5 of the status register is a "1" (ready state).

Figure 20. PROGRAM PAGE CACHE MODE Example



Notes:

1. See Note 3, Table 14, "PROGRAM/ERASE Characteristics" on page 20.
2. Check I/O[6:5] for internal ready/busy. Check I/O[1:0] for pass fail. RE# can stay LOW or pulse multiple times after a 70h command.

6.3 Internal Data Move

An internal data move requires two command sequences. Issue a READ for INTERNAL DATA MOVE (00h-35h) command first, then the PROGRAM for INTERNAL DATA MOVE (85h-10h) command. Data moves are only supported within the die from which data is read.

6.3.1 READ FOR INTERNAL DATA MOVE 00h-35h

The READ for INTERNAL DATA MOVE (00h-35h) command is used in conjunction with the PROGRAM for INTERNAL DATA MOVE (85h-10h) command. First (00h) is written to the command register, then the internal source address is written (five cycles). After the address is input, the READ for INTERNAL DATA MOVE (35h) command writes to the command register. This transfers a page from memory into the cache register.

The written column addresses are ignored even though all five ADDRESS cycles are required.

The memory device is now ready to accept the PROGRAM for INTERNAL DATA MOVE command. Please refer to the description of this command in the following section.

6.3.2 PROGRAM for INTERNAL DATA MOVE 85h-10h

After the READ for INTERNAL DATA MOVE (00h-35h) command has been issued and R/B# goes HIGH, the PROGRAM for INTERNAL DATA MOVE (85h-10h) command can be written to the command register. This command transfers the data from the cache register to the data register and programming of the new destination page begins. The sequence: 85h, destination address (five cycles), then 10h, is written to the device. After 10h is written, R/B# goes LOW while the control logic automatically programs the new page. The READ STATUS command can be used instead of the R/B# line to determine when the write is complete. When status register bit 6 = "1," bit 0 indicates if the operation was successful.

The RANDOM DATA INPUT (85h) command can be used during the PROGRAM for INTERNAL DATA MOVE command sequence to modify a word or multiple words of the original data. First, data is copied into the cache register using the 00h-35h command sequence, then the RANDOM DATA INPUT (85h) command is written along with the address of the data to be modified next. New data is input on the external data pins. This copies the new data into the cache register.

When 10h is written to the command register, the original data plus the modified data is transferred to the data register, and programming of the new page is started. The RANDOM DATA INPUT command can be issued as many times as necessary before starting the programming sequence with 10h.

Because INTERNAL DATA MOVE operations do not use external memory, ECC cannot be used to check for errors before programming the data to a new page. This can lead to a data error if the source page contains a bit error due to charge loss or charge gain. In the case that multiple INTERNAL DATA MOVE operations are performed, these bit errors may accumulate without correction. For this reason, it is highly recommended that systems using INTERNAL DATA MOVE operations also use a robust ECC scheme that can correct two or more bits per sector.

Figure 21. INTERNAL DATA MOVE

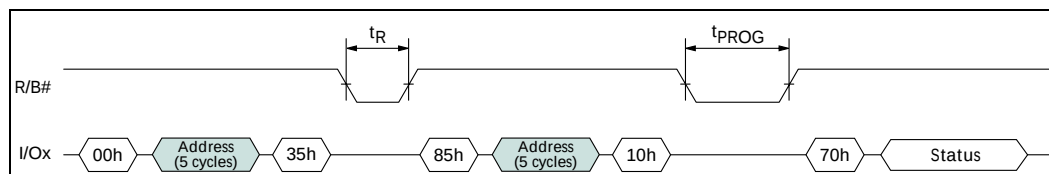
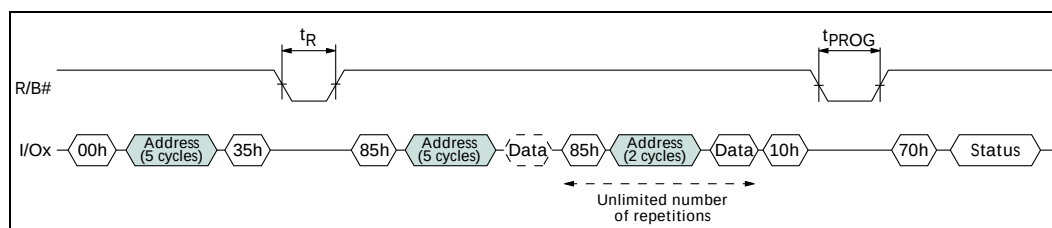



Figure 22. INTERNAL DATA MOVE with RANDOM DATA INPUT


6.4 BLOCK ERASE Operation

6.4.1 BLOCK ERASE 60h-D0h

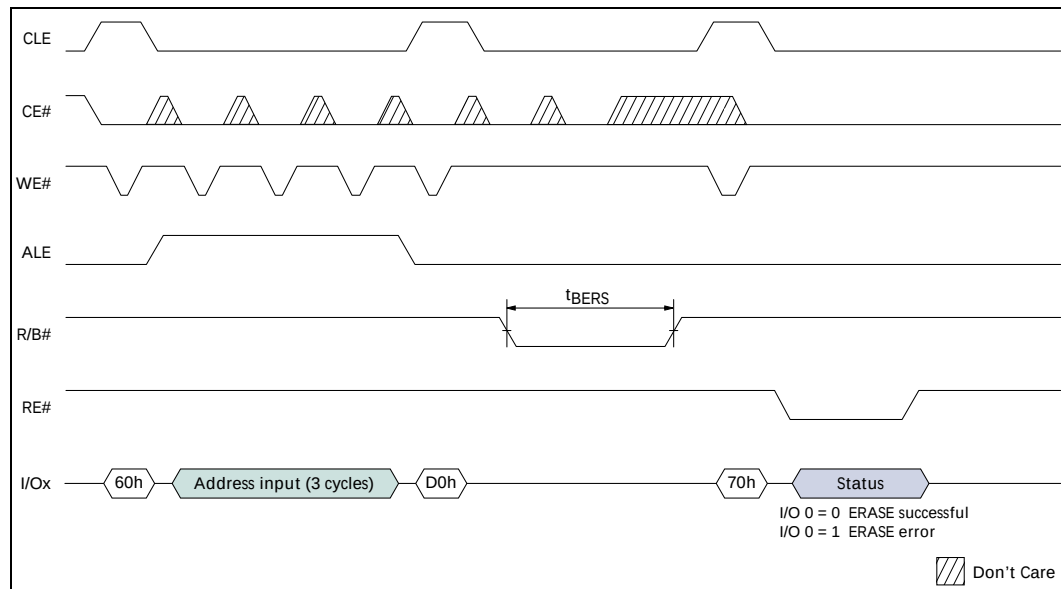
Erasing occurs at the block level. For example, the JS29F02G08AANB3 device has 2,048 erase blocks organized into 64 pages per block, 2,112 bytes per page (2,048 + 64 bytes). Each block is 132K bytes (128K + 4K bytes). The BLOCK ERASE command operates on one block at a time.

Three cycles of addresses BA[17:6] and PA[5:0] are required. Although page addresses PA[5:0] are loaded, they are "Don't Care" and are ignored for BLOCK ERASE operations. See [Table 2, "Intel® SS72 NAND Flash Memory Array Addressing \(x8\)"](#) on [page 10](#) for addressing details.

The actual command sequence is a two-step process. The ERASE SETUP (60h) command is first written to the command register. Then three cycles of addresses are written to the device. Next, the ERASE CONFIRM (D0h) command is written to the command register. At the rising edge of WE#, R/B# goes LOW and the control logic automatically controls the timing and erase-verify operations. R/B# stays LOW for the entire t_{BERS} erase time.

The READ STATUS (70h) command can be used to check the status of the BLOCK ERASE operation. When bit 6 = "1" the ERASE operation is complete. Bit 0 indicates a pass/fail condition where "0" = pass (see [Figure 17, "Status Register Operation"](#) on [page 31](#), and [Table 18, "Status Register Bit Definition"](#) on [page 31](#)).

Figure 23. BLOCK ERASE Operation



6.5 One Time Programmable (OTP) Area

This Intel® NAND Flash device offers a protected, one-time programmable NAND Flash memory area. Ten full pages (2,112 bytes or 1,056 words per page) of OTP data is available on the device, and the entire range is guaranteed to be good from the factory. The OTP area is accessible only through the OTP commands. Customers can use the OTP area any way they desire; typical uses include programming serial numbers or other data for permanent storage.

In Intel® NAND Flash devices, the OTP area leaves the factory in a non-written state (all bits are "1s"). Programming or partial-page programming enables the user to program only "0" bits in the OTP area. The OTP area cannot be erased, even if it is not protected. Protecting the OTP area simply prevents further programming of the OTP area.

While the OTP area is referred to as "one-time programmable," Intel provides a unique way to program and verify data—before permanently protecting it and preventing future changes.

OTP programming and protection are accomplished in two discrete operations. First, using the OTP DATA PROGRAM (A0h-10h) command, an OTP page is programmed entirely in one operation, or in up to four partial-page programming sequences. Second, the OTP area is permanently protected from further programming using the OTP DATA PROTECT (A5h-10h) command. The pages within the OTP area can always be read using the OTP DATA READ (AFh-30h) command, whether or not it is protected.

6.5.1 OTP DATA PROGRAM A0h-10h

The OTP DATA PROGRAM (A0h-10h) command is used to write data to the pages within the OTP area. An entire page can be programmed at one time, or the page can be partially programmed up to four times. There is no ERASE operation for the OTP pages.



The OTP DATA PROGRAM enables programming into an offset of an OTP page, using the two bytes of column address (CA[11:0]). The command is not compatible with the RANDOM DATA INPUT (85h) command. The OTP DATA PROGRAM command will not execute if the OTP area has been protected.

To use the OTP DATA PROGRAM command, issue the A0h command. Then issue five ADDRESS cycles: the first two ADDRESS cycles are the column address. For a $V_{CC}=1.7V-1.95V$ part the OTP page # is repeated in the third and fourth address cycles, and the fifth address cycle is 00h. For a $V_{CC}=2.7V-3.6V$ part the third cycle is the OTP page #, and the fourth and fifth address cycles are 00h-00h. See [Figure 24 on page 37](#) and [Figure 25 on page 38](#) for details.

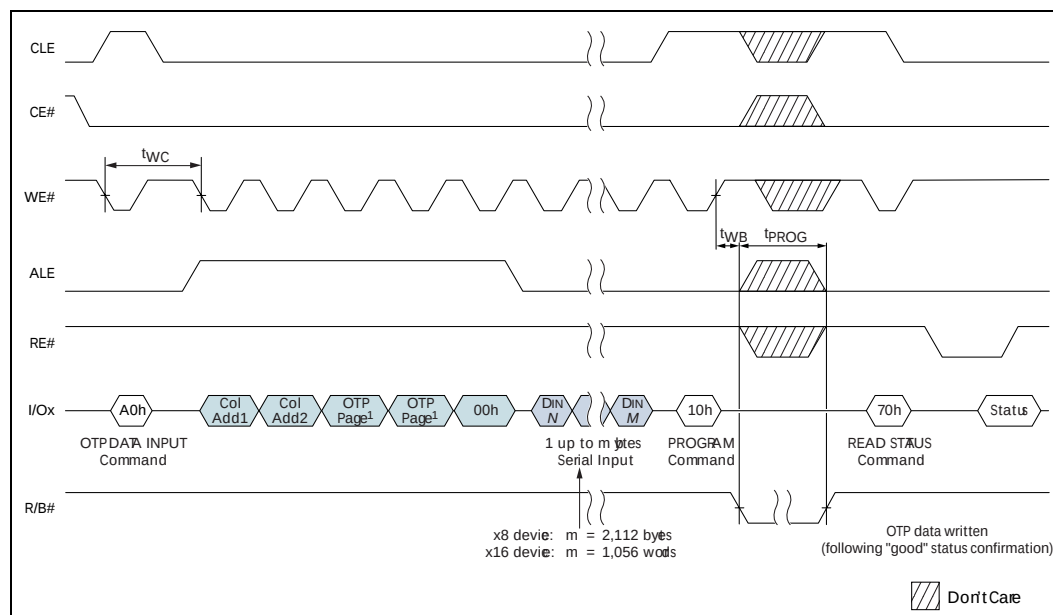
Next, write the data: from 1 to 2,112 bytes (x8 device), or from 1 to 1,056 words (x16 device). After data input is complete, issue the 10h command. The internal control logic automatically executes the proper programming algorithm and controls the necessary timing for programming and verification. Program verification only detects "1s" that are not successfully written to "0s."

R/B# goes LOW during the duration of the array programming time (t_{PROG}). The READ STATUS (70h) command is the only command valid during the OTP DATA PROGRAM operation. Bit 5 of the status register will reflect the state of R/B#. If bit 7 is "0," the OTP area has been protected; otherwise, it is not protected.

When the device is ready, read bit 0 of the status register to determine if the operation passed or failed (see [Table 18, "Status Register Bit Definition" on page 31](#)).

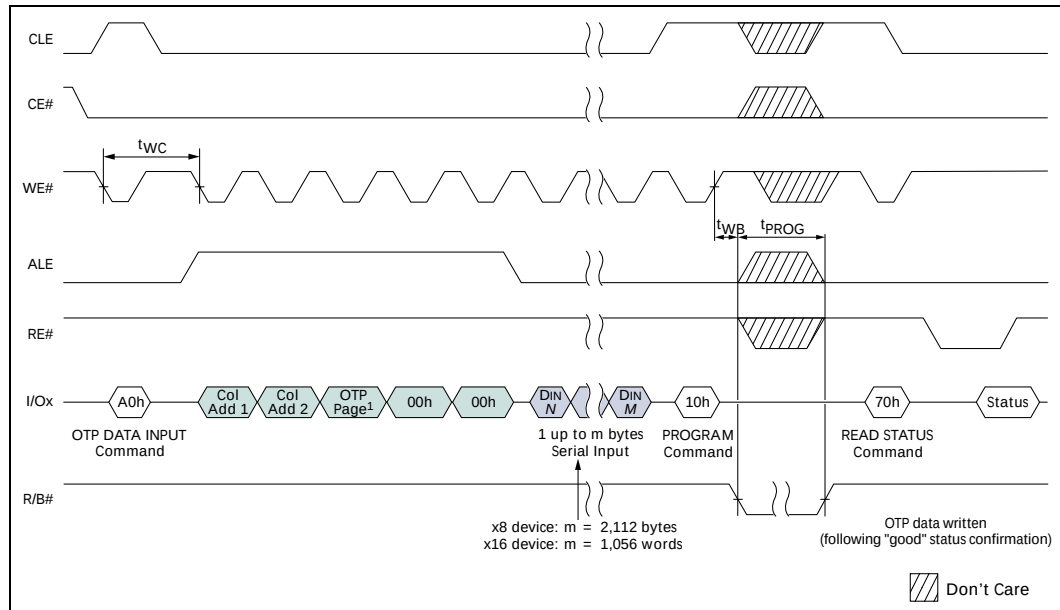
It is possible to program the OTP page a maximum of four times.

Figure 24. OTP DATA PROGRAM (1.8V Part)



Note: The OTP page must be within the 02h–0Bh range.

Figure 25. OTP DATA PROGRAM (3.3V Part)



Note: The OTP page must be within the 02h–0Bh range.

6.5.2 OTP DATA PROTECT A5h-10h

The OTP DATA PROTECT (A5h-10h) command is used to protect the data in the OTP area. After the data is protected it cannot be programmed further. When the OTP area is protected, the pages within the area are no longer programmable and cannot be unprotected.

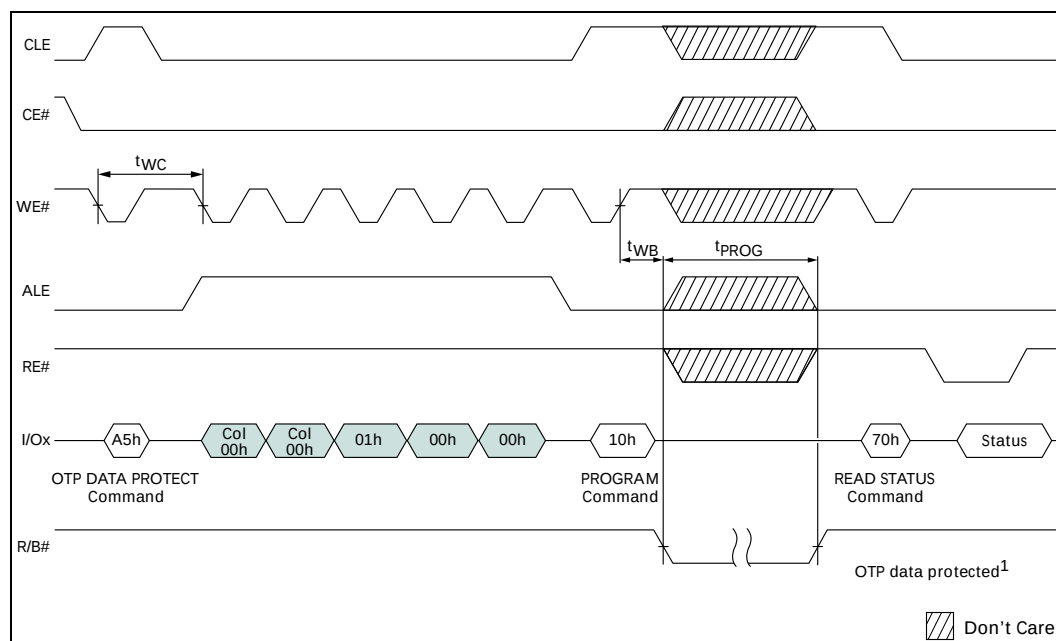
To use the OTP DATA PROTECT command, issue the A5h command. Next, issue the following five address cycles: 00h-00h-01h-00h-00h. Finally, issue the 10h command.

R/B# goes LOW while the OTP area is being protected. The protect command duration is similar to a normal page programming operation, t_{PROG} . The READ STATUS (70h) command is the only command valid during the OTP DATA PROTECT operation. Bit 5 of the status register will reflect the state of R/B#.

When the device is ready, read bit 0 of the status register to determine if the operation passed or failed (see Table 18, "Status Register Bit Definition" on page 31).



Figure 26. OTP DATA PROTECT



Note: OTP data is protected following "good" status confirmation.

6.5.3 OTP DATA READ AFh-30h

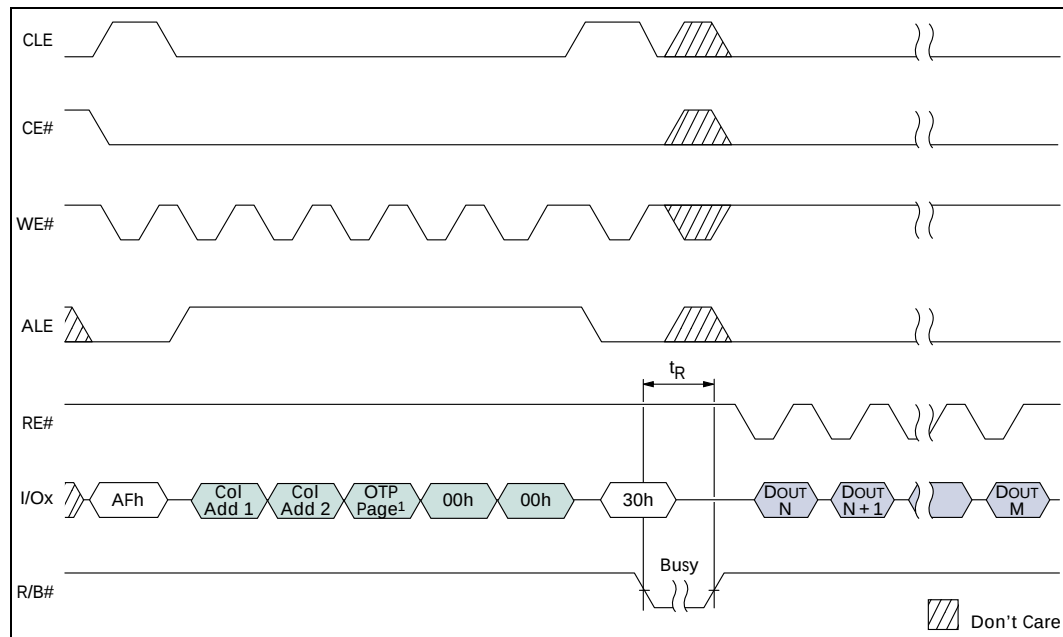
The OTP DATA READ (AFh-30h) command is used to read data from a page within the OTP area. An OTP page within the OTP area is available for reading data whether or not the area is protected.

To use the OTP DATA READ command, issue the AFh command. Then issue five ADDRESS cycles: the first two ADDRESS cycles are the column address, and for the remaining three cycles select a page in the range of 02h-00h-00h through 0Bh-00h-00h. Finally, issue the 30h command.

R/B# goes LOW (t_R) while the data is moved from the OTP page to the data register. The READ STATUS (70h) command and the RESET (FFh) command are the only commands valid during the OTP DATA READ operation. Bit 5 of the status register will reflect the state of R/B#. For details, refer to [Table 18, "Status Register Bit Definition" on page 31](#).

Normal READ operation timings apply to OTP read accesses. Additional pages within the OTP area can be selected by repeating the OTP DATA READ command.

Figure 27. OTP DATA READ



Note: The OTP page must be within the 02h–0Bh range.

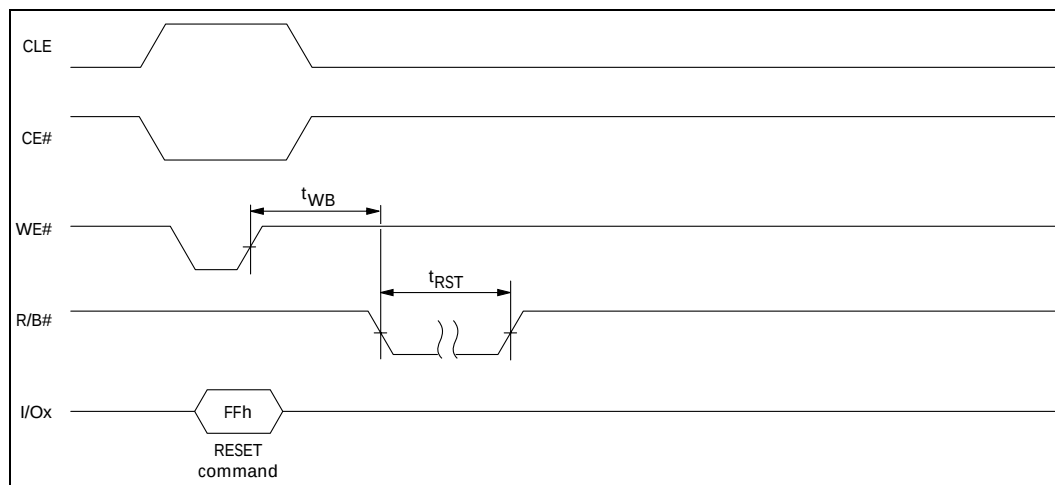
6.6 RESET Operation

6.6.1 RESET FFh

The RESET command is used to put the memory device into a known condition and to abort a command sequence in progress.

READ, PROGRAM, and ERASE commands can be aborted while the device is in the busy state. The contents of the memory location being programmed or the block being erased are no longer valid. The data may be partially erased or programmed, and is invalid. The command register is cleared and is ready for the next command. The data register and cache register contents are marked invalid.

The status register contains the value E0h when WP# is HIGH; otherwise it is written with a 60h value. R/B# goes low for t_{RST} after the RESET command is written to the command register.

Figure 28. RESET Operation**Table 19. Status Register Contents After RESET Operation**

Condition	Status	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Hex
WP# HIGH	Ready	1	1	1	0	0	0	0	0	E0h
WP# LOW	Ready and write protected	0	1	1	0	0	0	0	0	60h

6.7 WRITE PROTECT Operation

It is possible to enable and disable PROGRAM and ERASE commands using the WP# pin. The following figures illustrate the setup time (t_{WW}) required from WP# toggling until a PROGRAM or ERASE command is latched into the command register. After command cycle 1 is latched, the WP# pin must not be toggled until the command is complete and the device is ready (status register bit 5 is "1").

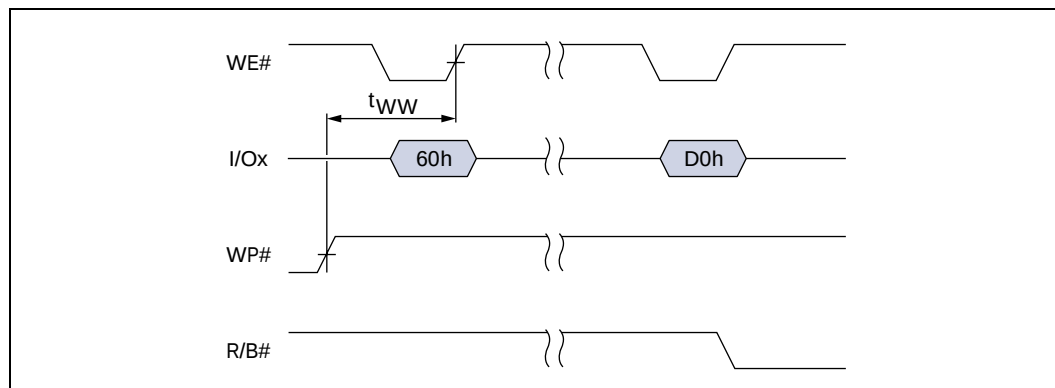
Figure 29. ERASE Enable

Figure 30. ERASE Disable

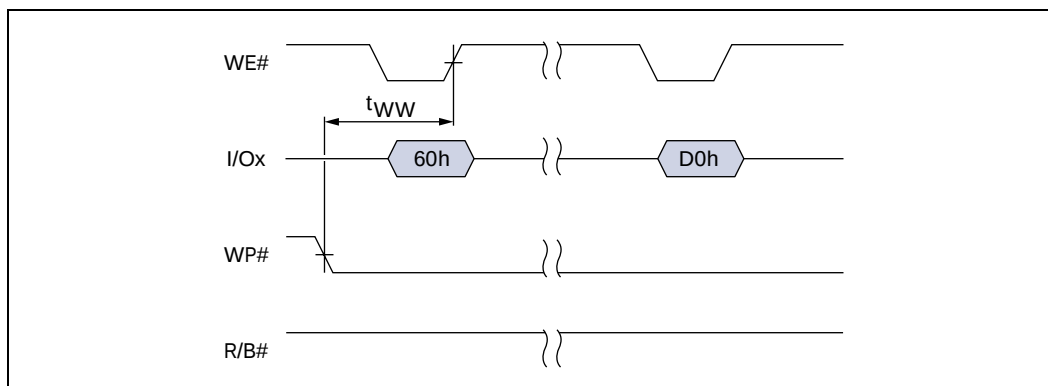


Figure 31. PROGRAM Enable

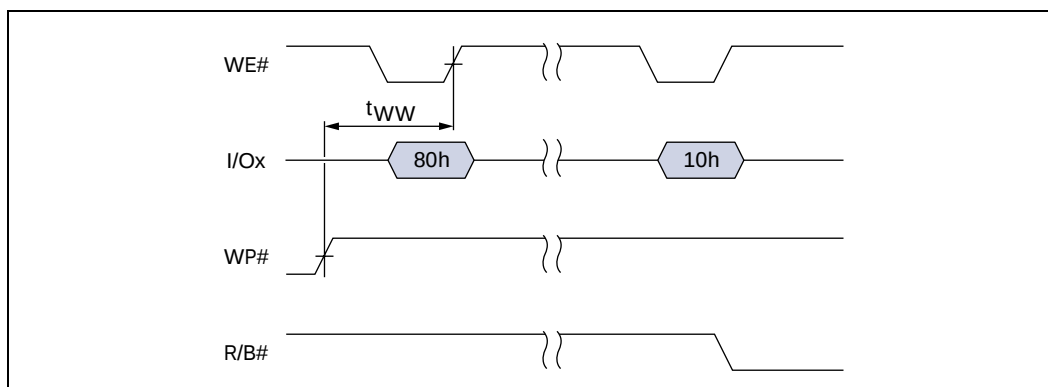
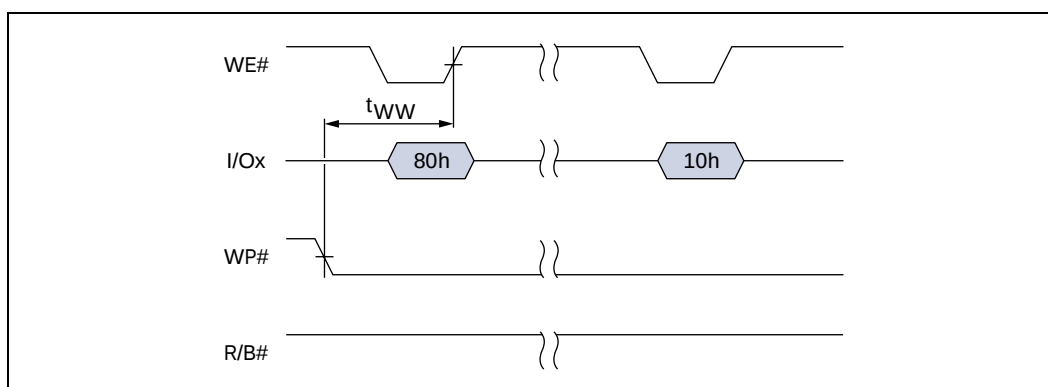


Figure 32. PROGRAM Disable



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7.0 Error Management

Intel® NAND Flash devices are specified to have a minimum of 2,008 (NVB) valid blocks out of every 2,048 total available blocks. This means the devices may have blocks that are invalid when they are shipped. An invalid block is one that contains one or more bad bits. Additional bad blocks may develop with use. However, the total number of available blocks will not fall below NVB during the endurance life of the product.

Although NAND Flash memory devices may contain bad blocks, they can be used quite reliably in systems that provide bad-block mapping, bad-block replacement, and error correction algorithms. This type of software environment ensures data integrity.

Internal circuitry isolates each block from other blocks, so the presence of a bad block does not affect the operation of the rest of the Flash device.

The first block (physical block address 00h) for each CE# is guaranteed to be free of defects (up to 1,000 PROGRAM/ERASE cycles) when shipped from the factory. This provides a reliable location for storing boot code and critical boot information.

Before NAND Flash devices are shipped from Intel, they are erased. The factory identifies invalid blocks before shipping by programming data other than FFh (x8) or FFFFh (x16) into the first spare location (column address 2,048 for x8 devices, or column address 1,024 for x16 devices) of the first or second page of each bad block.

System software should check the first spare address on the first two pages of each block prior to performing any erase or programming operations on the NAND Flash device. A bad block table can then be created, allowing system software to map around these areas. Factory testing is performed under worst-case conditions. Because blocks marked "bad" may be marginal, it may not be possible to recover this information if the block is erased.

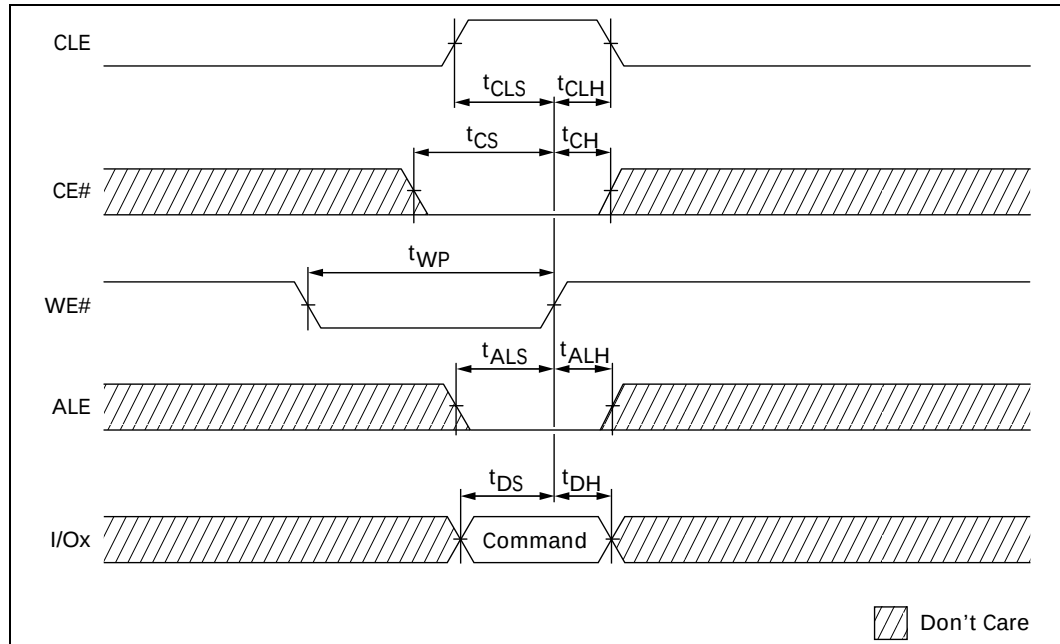
Over time, some memory locations may fail to program or erase properly. In order to ensure that data is stored properly over the life of the NAND Flash device, certain precautions must be taken, such as:

- Always check status after a PROGRAM, ERASE, or DATA MOVE operation.
- Use some type of error detection and correction algorithm to recover from single-bit errors per 528 bytes of data.
- Use a bad-block replacement algorithm.



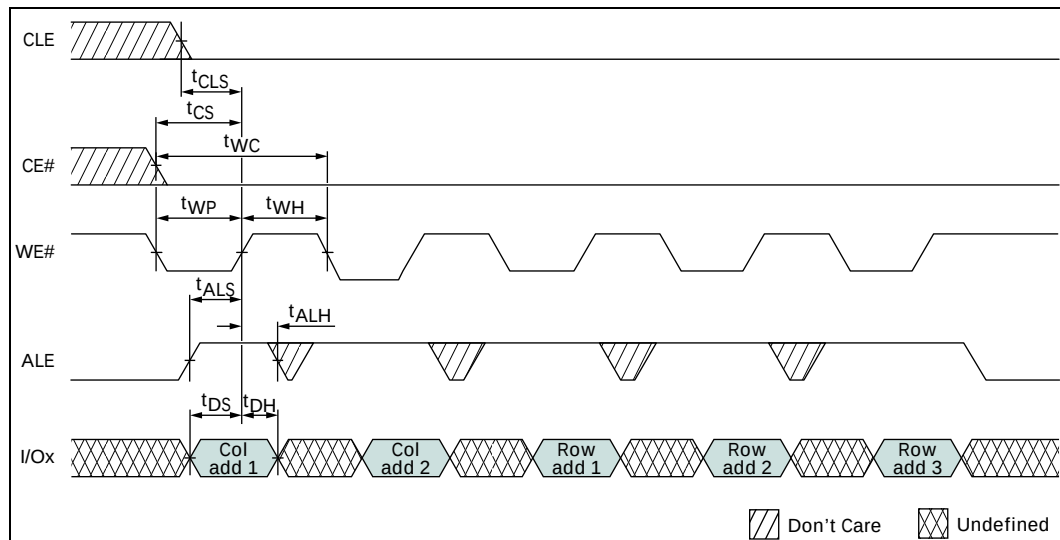
8.0 Timing Diagrams

Figure 33. COMMAND LATCH Cycle



Note: x16: I/O[15:8] must be set to "0."

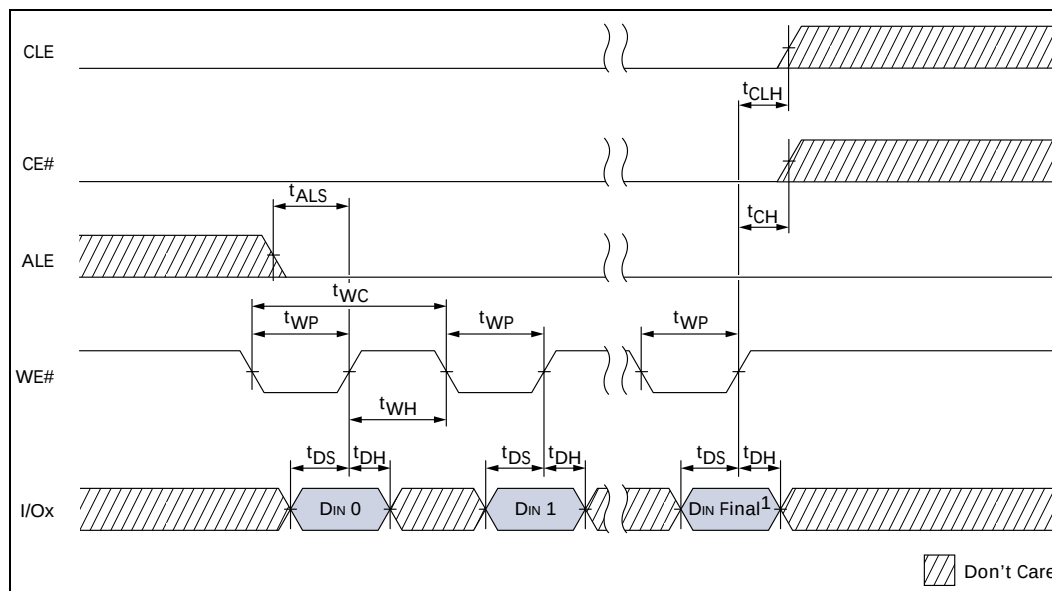
Figure 34. ADDRESS LATCH Cycle



Note: x16: I/O [15:8] must be set to "0."



Figure 35. INPUT DATA LATCH



Note: $D_{IN} \text{ Final} = 2,111 \text{ (x8) or } 1,055 \text{ (x16)}$.

Figure 36. SERIAL ACCESS Cycle After READ

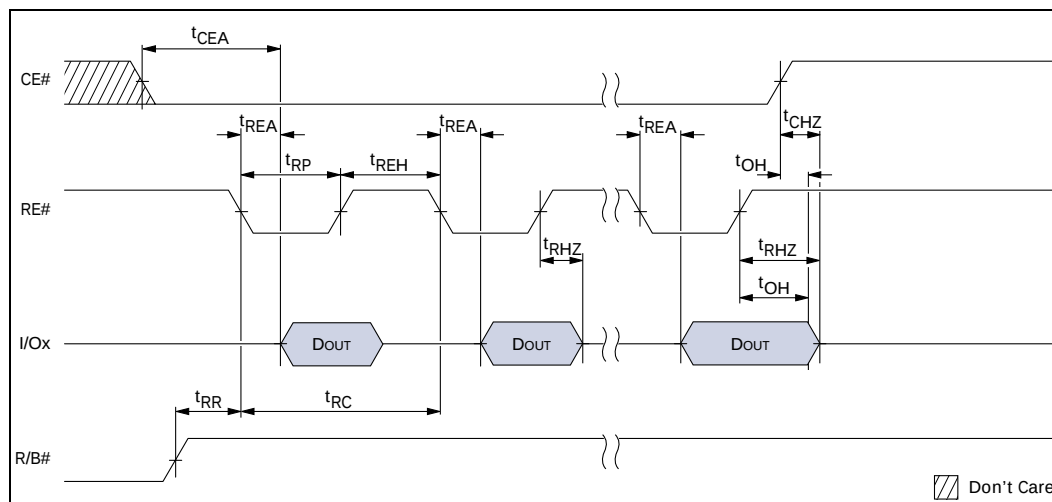


Figure 37. READ STATUS Cycle

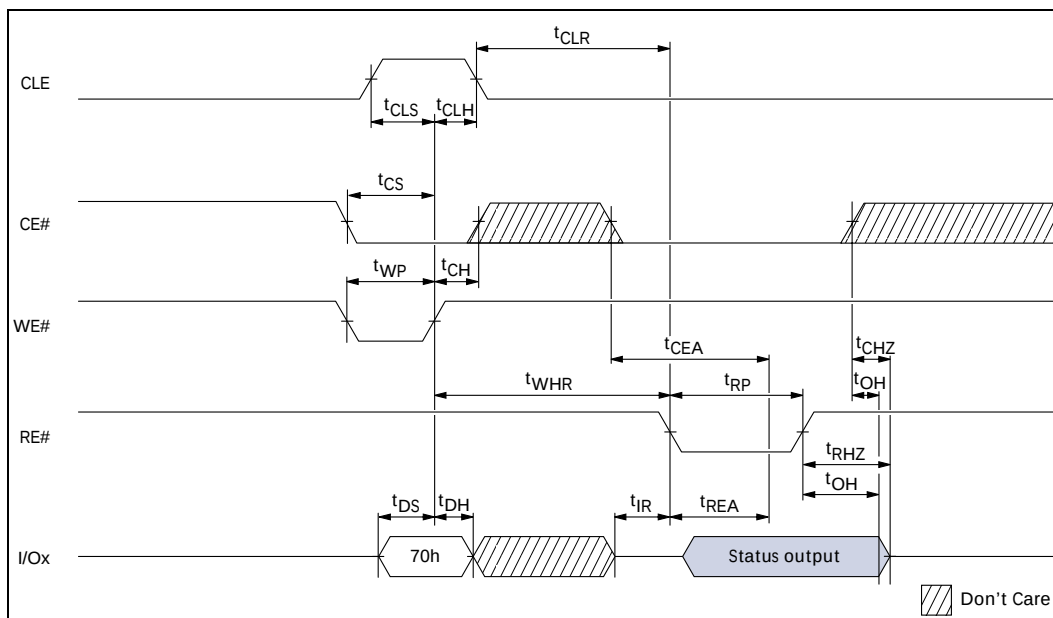


Figure 38. PAGE READ

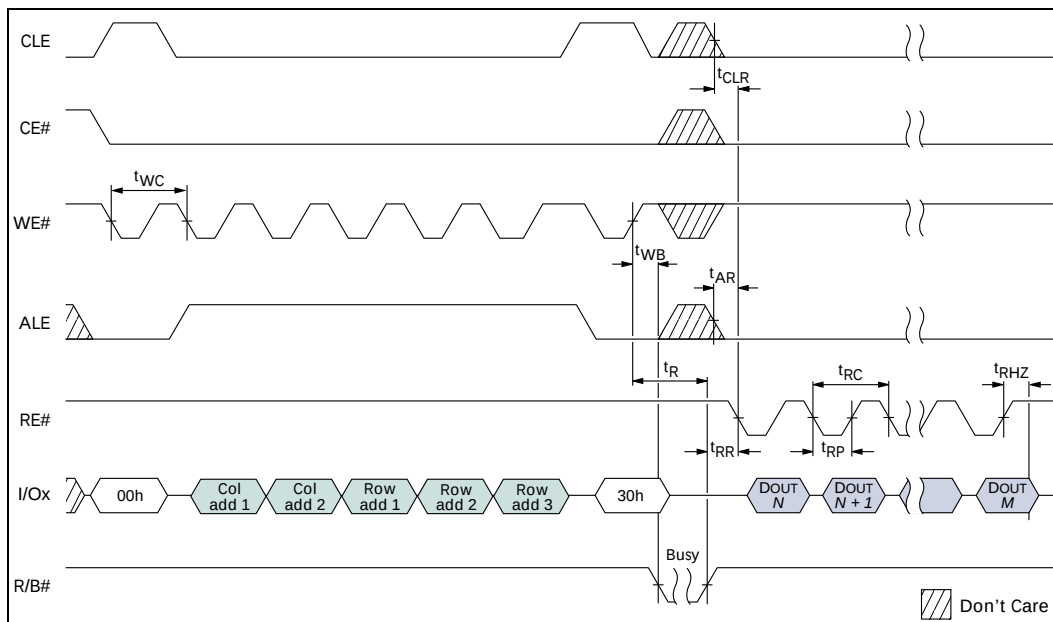


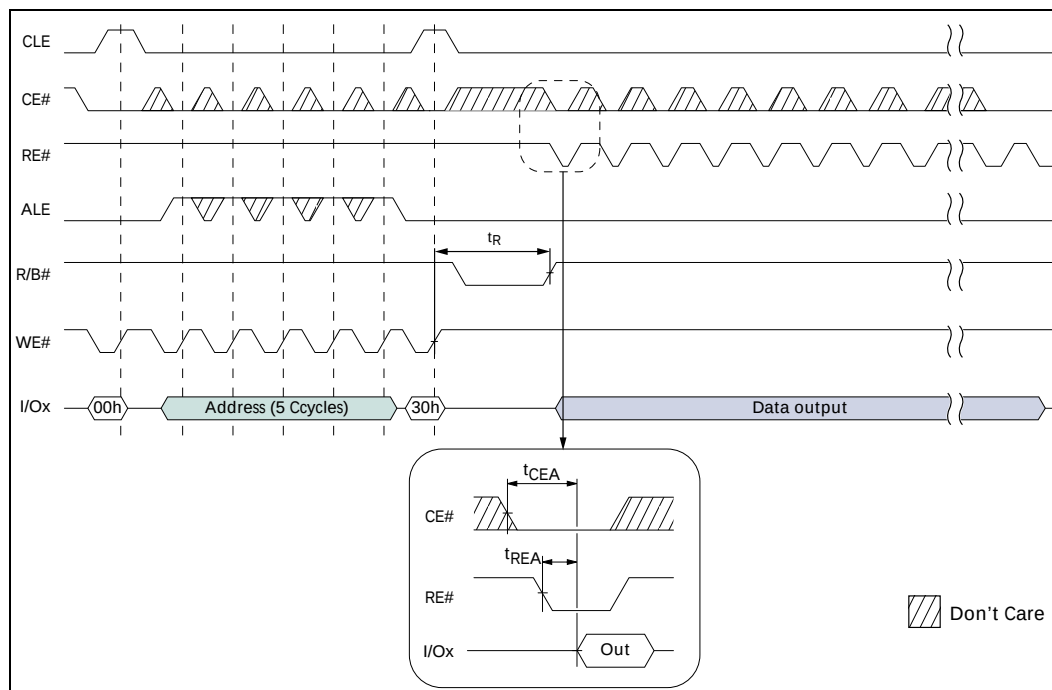
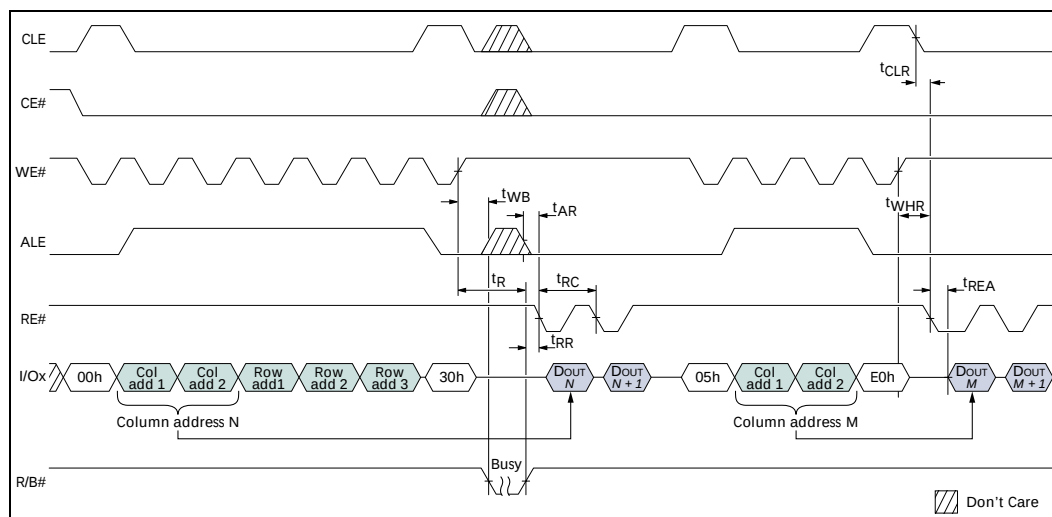

Figure 39. PAGE READ Operation with CE# Don't Care

Figure 40. RANDOM DATA READ


Figure 41. PAGE READ CACHE MODE Timing Diagram (Part 1 of 2)

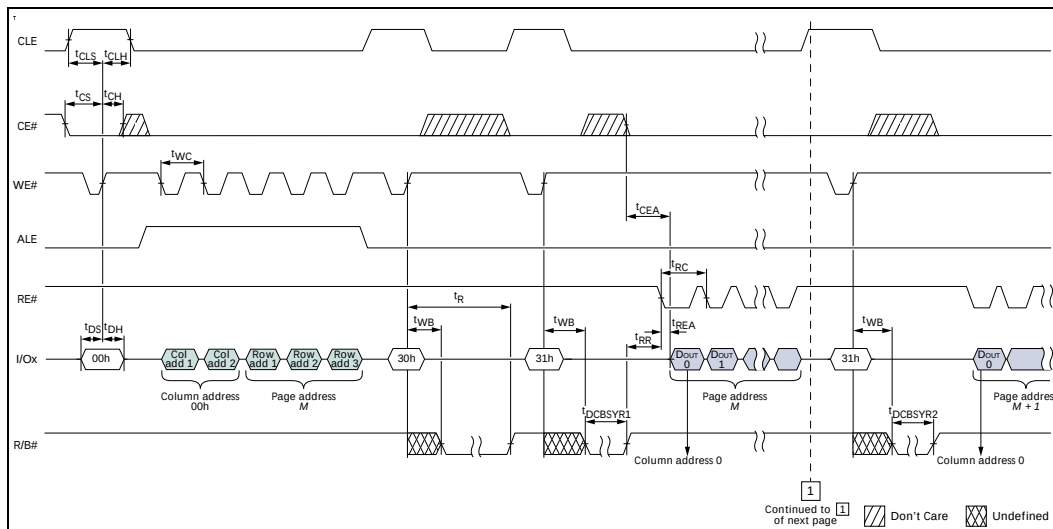


Figure 42. PAGE READ CACHE MODE Timing Diagram (Part 2 of 2)

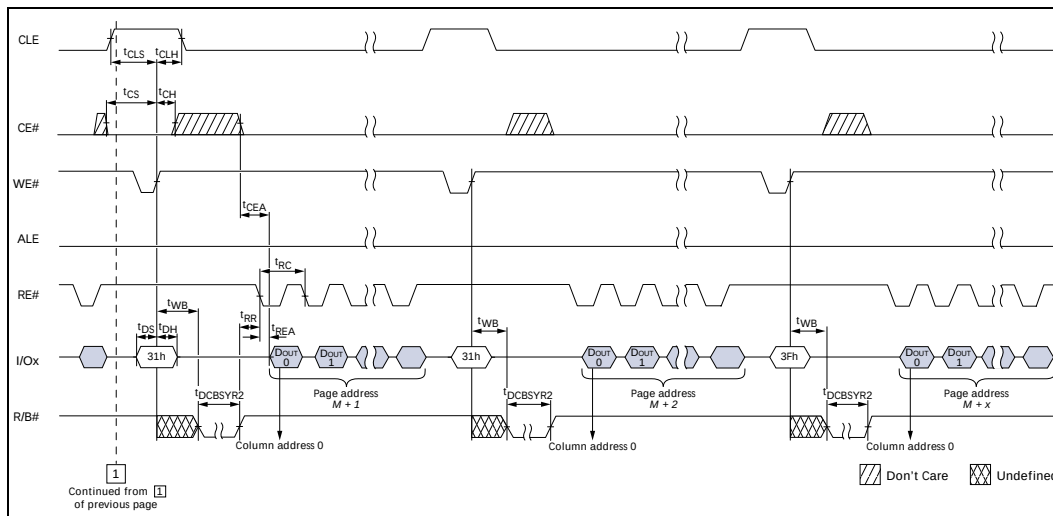




Figure 43. PAGE READ CACHE MODE Timing without R/B# (Part 1 of 2)

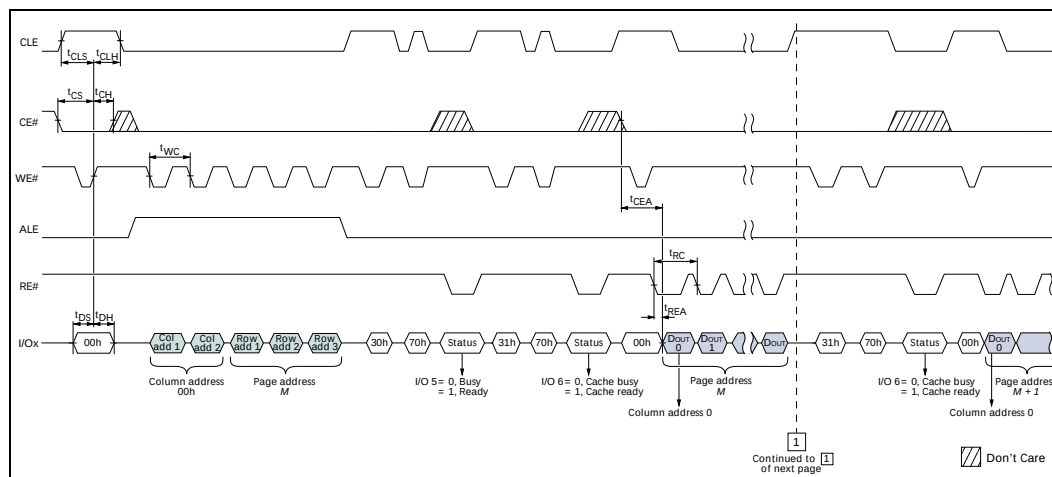


Figure 44. PAGE READ CACHE MODE Timing without R/B# (Part 2 of 2)

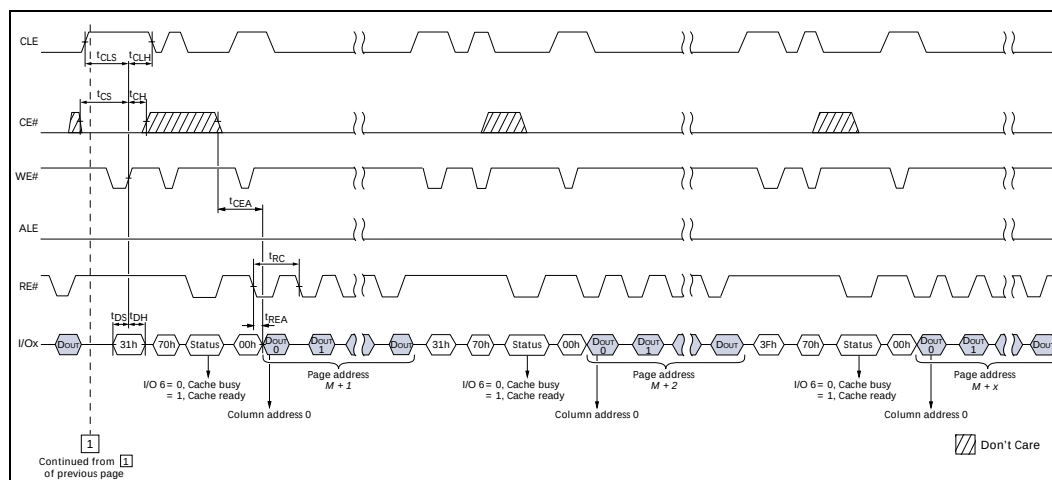
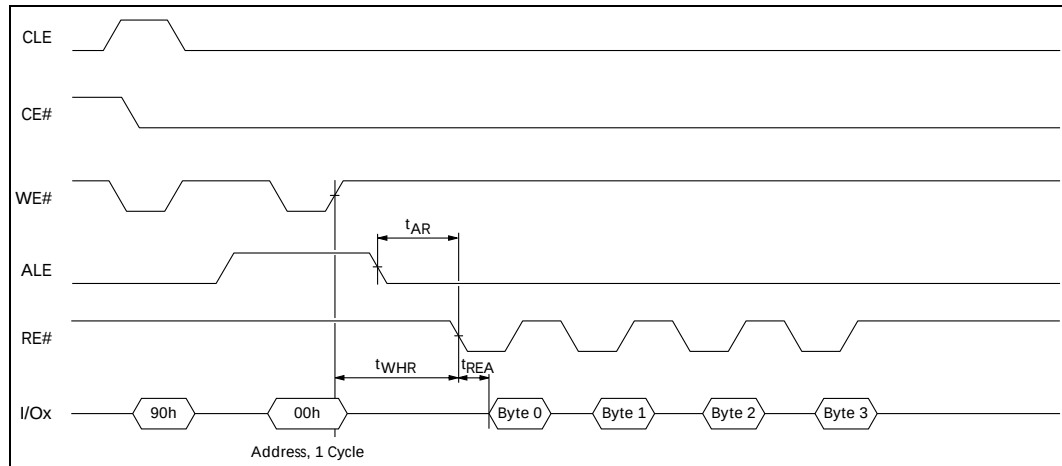


Figure 45. READ ID Operation



Note: See Table 17, "Device ID and Configuration Codes" on page 30.

Figure 46. PROGRAM PAGE Operation

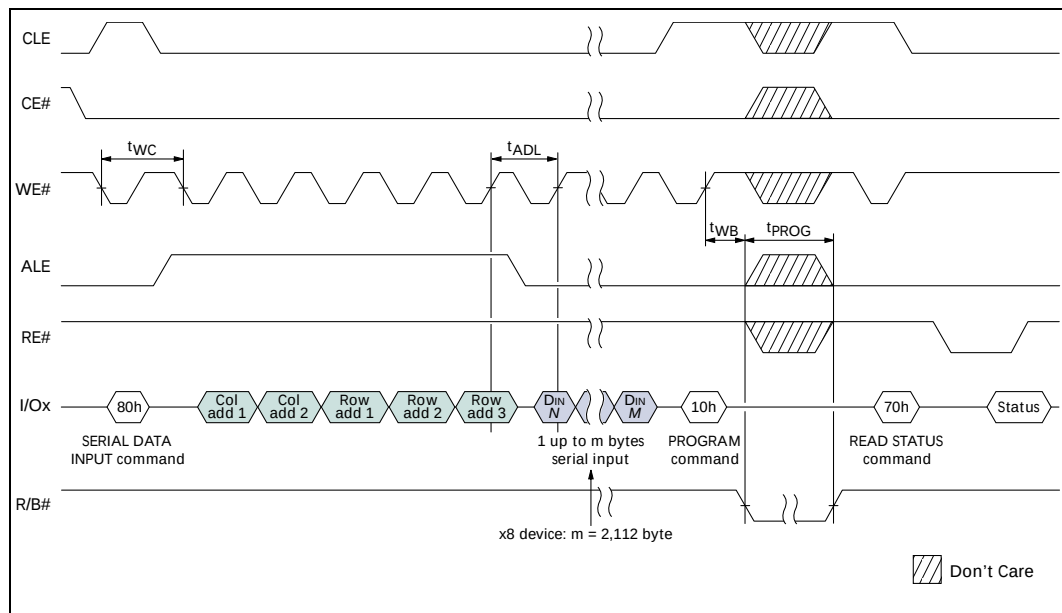


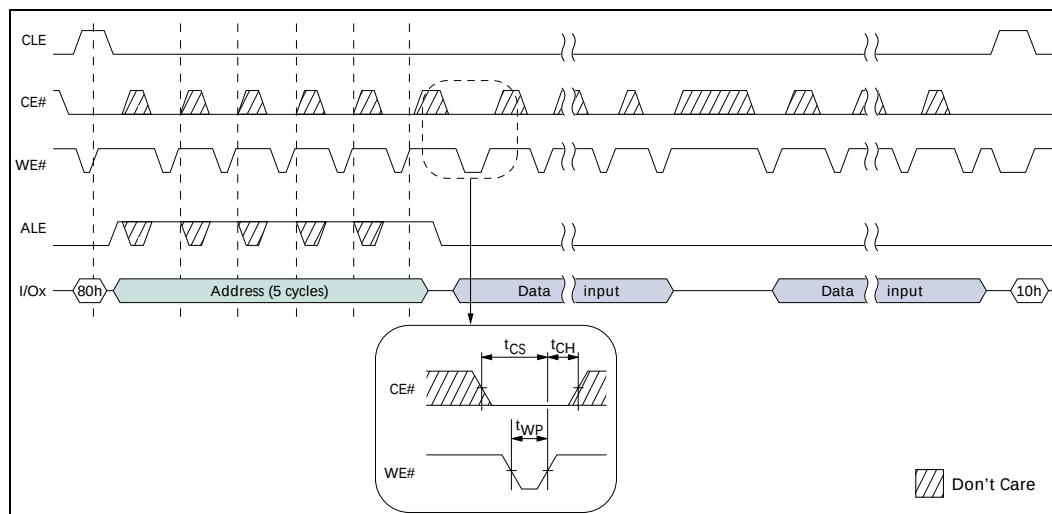
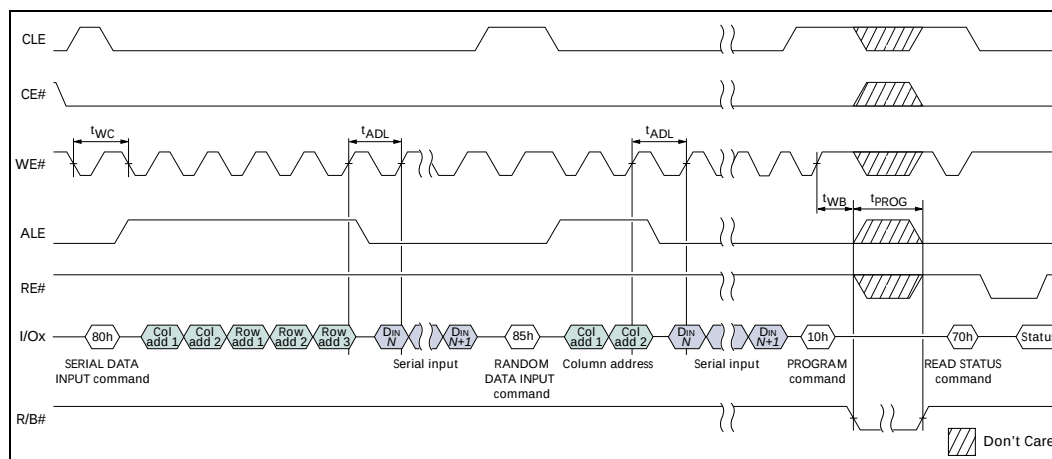
Figure 47. PROGRAM PAGE Operation with CE# Don't Care**Figure 48. PROGRAM PAGE Operation with RANDOM DATA INPUT**

Figure 49. INTERNAL DATA MOVE

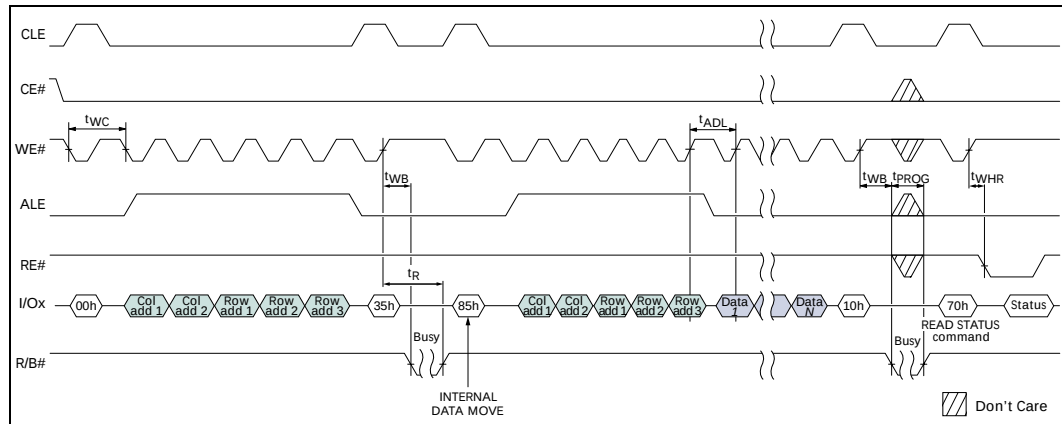


Figure 50. PROGRAM PAGE CACHE MODE

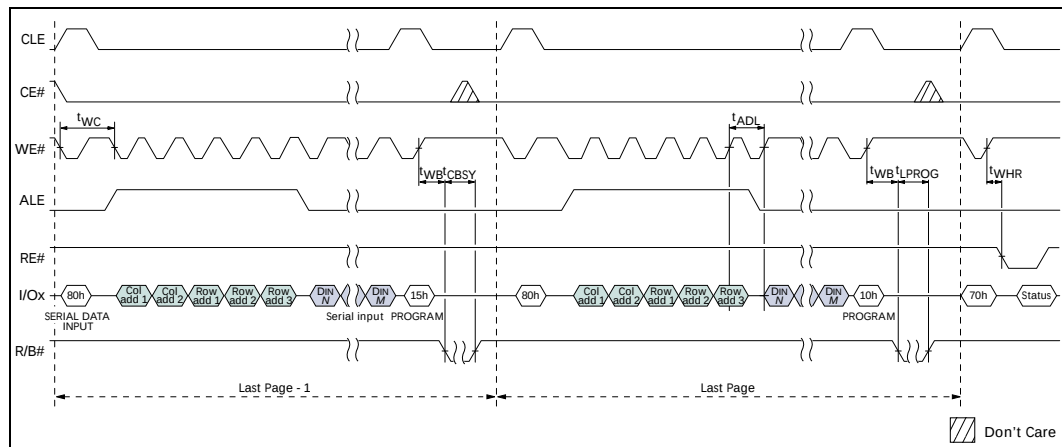


Figure 51. PROGRAM PAGE CACHE MODE Ending on 15h

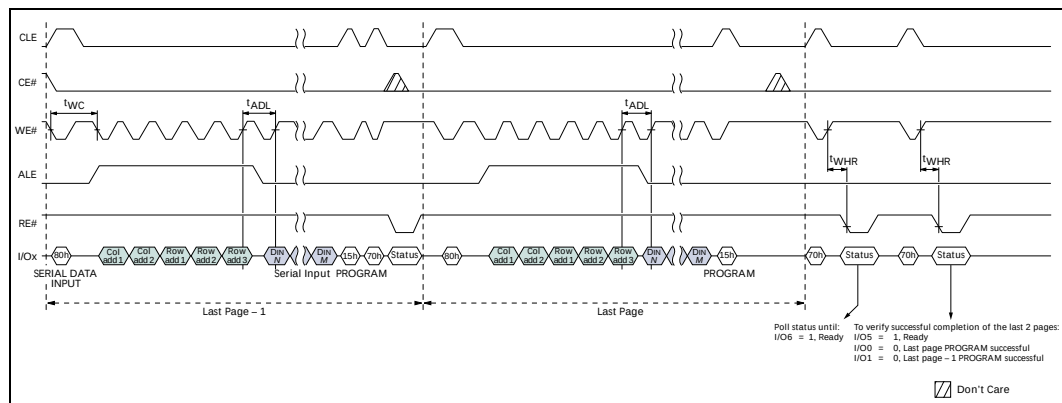
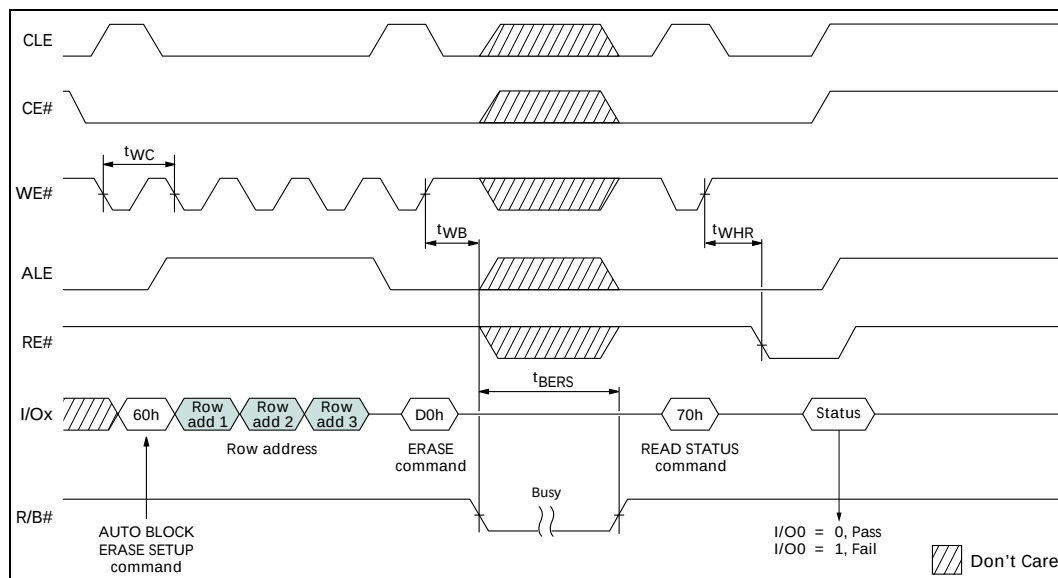


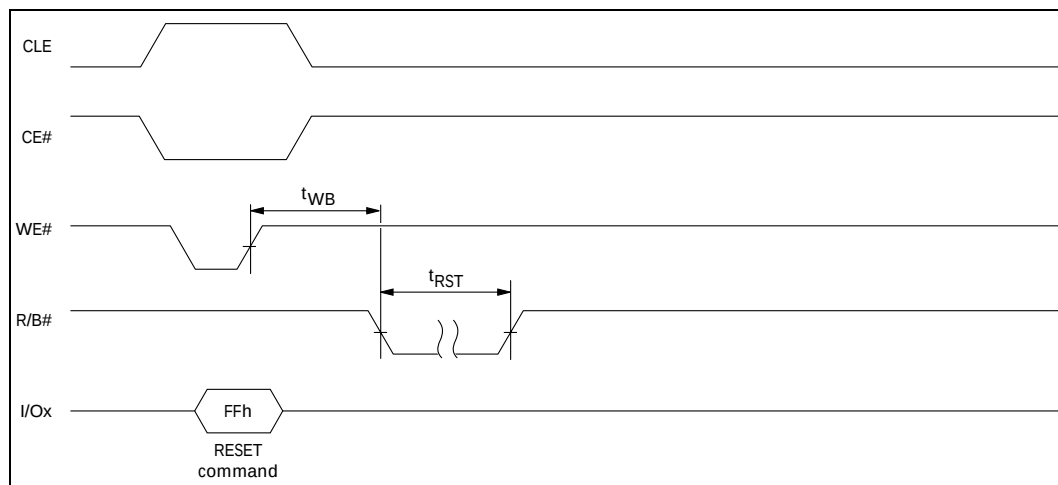


Figure 52. BLOCK ERASE Operation

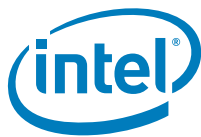


Note: See Table 17, "Device ID and Configuration Codes" on page 30 for actual values.

Figure 53. RESET Operation



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9.0 Ordering Information

Figure 54, “Decoder” on page 54 provides the device part number decoder and Table 20, “Intel® NAND Flash Memory Ordering Information” on page 54 provides the available combinations. For combinations not listed, please contact your local Intel sales office.

Figure 54. Decoder

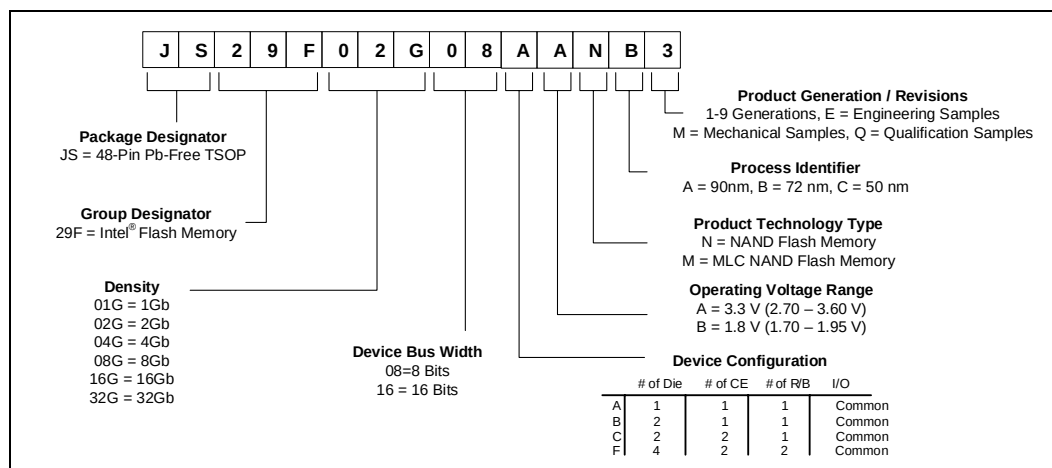


Table 20. Intel® NAND Flash Memory Ordering Information

IM L1 Part Number	Marketing Device # (1st Mark Line)	MM # (2nd Mark Line)	Device Nomenclature
JS29F02G08AANB3	29F02G08AANB3	880202	2Gb, x8, 1 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (1000pc T&R), Pb-Free)
		880203	2Gb, x8, 1 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (11 Tray Pack), Pb-Free)
JS29F04G08BANB3	29F04G08BANB3	884090	2Gb, x8, 2 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (1000pc T&R), Pb-Free)
		884116	2Gb, x8, 2 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (11 Tray Pack), Pb-Free)
JS29F08G08FANB3	29F08G08FANB3	880197	2Gb, x8, 4 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (1000pc T&R), Pb-Free)
		880198	2Gb, x8, 4 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (11 Tray Pack), Pb-Free)
TBD	TBD	TBD	2Gb, x16, 1 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (1000pc T&R), Pb-Free)
			2Gb, x16, 1 die, 3.3 V, NAND, 72 nm, 3rd Gen Intel Si (11 Tray Pack), Pb-Free)

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